How maimai DX Works

The Game, the Cabinet and the Rules of Version 1.60 (CiRCLE)

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Abstract

maimai DX is SEGA's arcade rhythm game. It is played on a round touch screen with eight buttons around its edge. This paper explains how version 1.60 (CiRCLE) works, from the cabinet hardware to the number on the result screen.

It covers the cabinet, with its portrait display, eight lit buttons, 34-sensor touch panel, cameras and card reader. It covers what a song brings: up to six charts, a music video and its sounds. It shows how a chart is built on a grid of 384 steps per bar from taps, holds, slides, touches and breaks. It then gives the scoring rules with their real numbers and formulas: the timing windows, the points per judgement, the 101% achievement, the DX score, the medals, the fourteen ranks and the rating. It walks through a credit screen by screen. Finally it describes the ALL.Net server infrastructure, and how a cabinet signs in, reads a card and loads and saves a player's profile. A closing reference lists every formula the game uses, each with a worked example.

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1. Introduction

In brief. maimai DX is SEGA's arcade rhythm game, played on a round touch screen ringed by eight buttons. This paper explains how version 1.60 (CiRCLE) works: the cabinet, the charts, the scoring rules, a credit from start to finish, and the network behind it. Every rule comes with its real numbers, and every score comes with its formula.

1.1 The game in one page

The cabinet looks like a front-loading washing machine. A large round window holds a touch screen. Eight physical buttons sit around the window's edge. One cabinet holds two of these stations side by side, so two people can play together.

Notes appear near the centre of the screen and travel outward to a judgement ring. The ring runs through the eight buttons. The player hits each note as it reaches the ring, either by pressing the button on its lane or by touching the screen where it lands. Some notes are slides, which the player traces across the glass with a finger. A second display area above the round screen shows the song, the score, the timer and the player's partner character.

1.2 How a play works

A credit buys a fixed number of songs, called tracks, usually three. A credit runs like this:

1. The player signs in with an Aime card, or plays as a guest.
2. The player picks options, then a song and a difficulty.
3. The song plays with its music video behind the field. The game judges every note.
4. The result screen shows the score, the rank, the medals and the change in rating.
5. Steps 2 to 4 repeat for each track. At the end the cabinet saves everything to the network.

The main score is called *achievement*. Hitting every note perfectly gives 100%. Break notes can add up to 1% more, so the best possible play reads 101%. Achievement sets a letter *rank* from D to SSS+. A second score, the *DX score*, gives up to 3 points per note. The player's overall number is the *rating*, built from the player's 50 best charts.

1.3 Terms

Table 1. Terms used throughout the paper.

Term	Meaning
Chart	One difficulty of one song. A song has up to six: BASIC, ADVANCED, EXPERT, MASTER, Re:MASTER and an optional novelty chart called utage
Lane	One of the eight directions from the centre to a button, numbered 1 to 8 clockwise from the top right
Sensor	One of the 34 touch zones of the screen, in five groups named A to E
Achievement	The percentage score of a play, from 0 to 101.0000%
DX score	A second score worth 0 to 3 points per note
Rank	The letter grade an achievement earns, from D to SSS+
Rating	A player's overall number: the sum of the ratings of their 50 best charts
Credit	One paid or free session of several tracks
Partner	The character who keeps the player company on screen and talks during the credit
ALL.Net	SEGA's network service that every cabinet connects to
Aime	SEGA's player card. It identifies a player across all SEGA games

1.4 How this paper is organised

- **Section 2, The cabinet.** Screens, buttons, touch panel, lights, cameras, card reader and test mode.
- **Section 3, Songs, charts and media.** What a song brings: its charts, music video and sounds.
- **Section 4, Charts.** The timing grid, every note type, the slide shapes and the touch sensors.
- **Section 5, Judgement and scoring.** Timing windows, points, achievement, medals, ranks and rating.
- **Section 6, A credit, screen by screen.** Every screen with its timer, plus options, collections and partners.
- **Section 7, The cabinet online.** The ALL.Net server infrastructure, the Aime card and how a profile is loaded and saved.
- **Section 8, Conclusion.**
- **Section 9, Formula reference.** Every formula the game uses for a score, each with a worked example.

2. The cabinet

In brief. This section describes the maimai DX cabinet as the game sees it: one portrait screen per player, eight ring buttons, a 34-sensor touch panel, lighting, cameras and a card reader. Two programs share the cabinet PC. The game drives the screen and the devices close to play, while a common SEGA service handles coins, cards and the network.

2.1 Overview

A maimai DX cabinet seats two players side by side. Each player faces a portrait screen behind a round glass window. Eight buttons sit in the ring around the window. A touch panel covers the glass. Each side also has a card reader for the Aime card, the player's membership card. One player camera above the cabinet serves both sides.

The game is a Unity application on a Windows PC inside the cabinet. It shares the PC with the cabinet service, a SEGA program common to its arcade titles. Figure 1 shows which devices belong to which program.

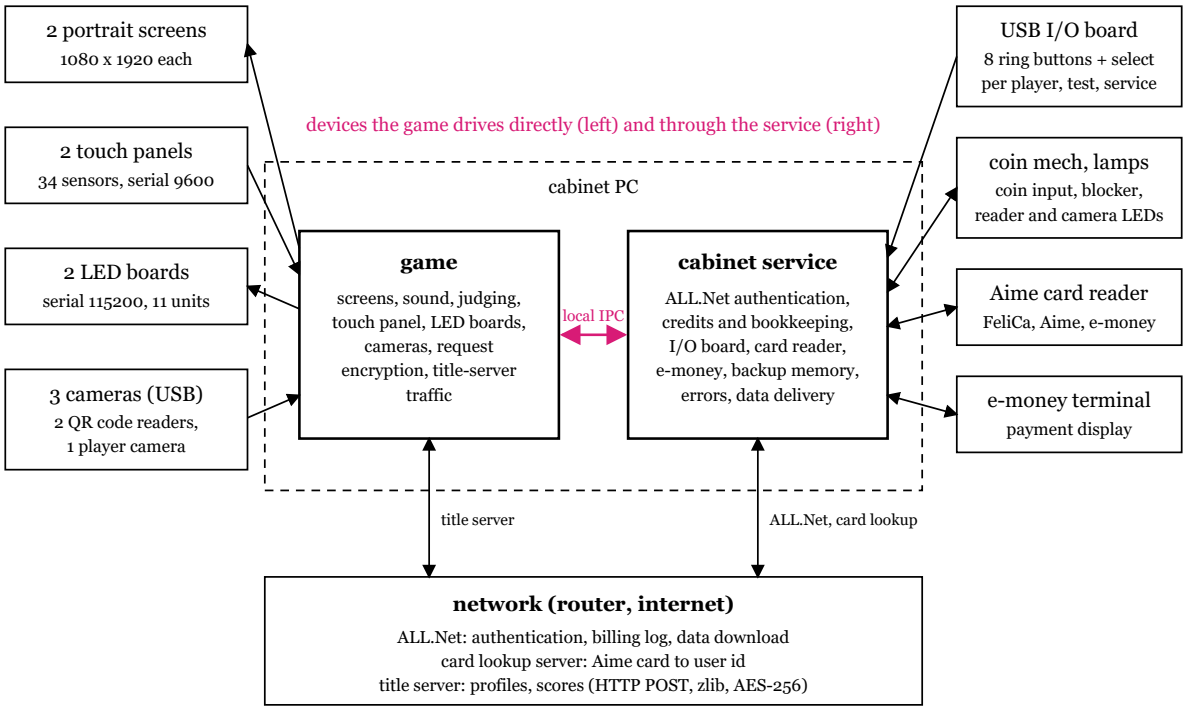


Figure 1. Cabinet architecture: the game and the cabinet service run side by side on the cabinet PC and talk over a local channel; the screens, touch panels, LED boards and cameras belong to the game, while the button I/O board, coin mechanism, card reader and e-money terminal belong to the service.

2.2 The screen

Each player has a 1920×1080 monitor mounted on its side, turned by 270°. The player therefore sees a portrait image 1080 pixels wide and 1920 pixels tall. The game draws both sides as two

independent screens with the same layout.

Every screen splits this portrait area into two parts, as Figure 2 shows. The lower square is the **play field**, where notes travel and are judged. The upper strip is the **sub-monitor**, which shows the player's name plate, rating, partner character, timers and messages. The game places both areas in layout units, with the origin at the screen centre and y pointing up.

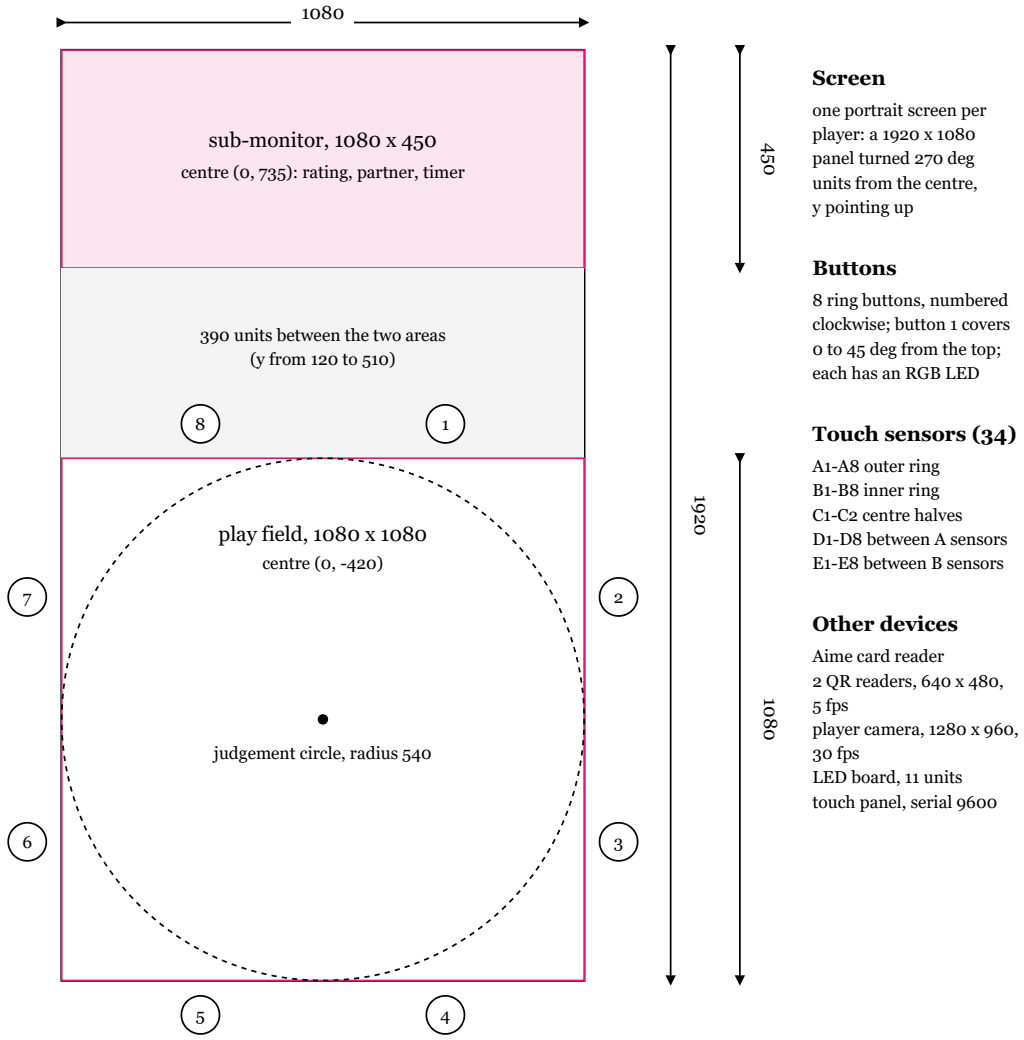
Table 2. The two areas of the portrait screen, in layout units.

Area	Size	Centre	Vertical span
Play field	1080 × 1080	(0, −420)	−960 to 120
Sub-monitor	1080 × 450	(0, 735)	510 to 960
Band between them	1080 × 390	(0, 315)	120 to 510

The judgement circle has radius 540, so it is inscribed in the play field. Its centre sits 420 units below the screen centre:

$$\text{field span} = -420 \pm 540 = [-960,\ 120]$$

The 390-unit band between the two areas carries no game content. On the cabinet it lies behind the bezel between the round window and the upper window.



Scale 0.28 px per unit. Button positions are schematic: outside the circle, at the lane angle. The physical button radius is not part of the screen layout.

Figure 2. The portrait screen of one player: the 1080×1080 play field with the judgement circle, the 1080×450 sub-monitor above it, the eight ring buttons, and the other devices of one side.

2.3 Buttons

The eight ring buttons are numbered clockwise from the top. Button 1 covers the 45° sector just right of twelve o'clock, and button 8 the sector just left of it. Each player also has a Select button. The Test and Service buttons for the operator sit behind the coin door.

A USB I/O board reads all 20 switches. The cabinet service owns this board and publishes the switch states. The game samples them once per display frame. For each switch it records whether the switch is down, and whether it went down or up in this frame. It also ignores a switch that bounces back on one frame after a release.

The same board drives five simple outputs:

- the coin blocker, which stops coins from entering;
- the LEDs of the left and right card readers;
- a ring light and a red recording light on the player camera.

2.4 The touch panel

The touch panel is a serial device, not a Windows touch screen. Each side has its own panel controller on a serial line at 9600 baud. The panel divides the screen into 34 sensors in five groups.

Table 3. The touch sensor groups.

Group	Count	Position
A	8	outer edge of the circle, one per lane
B	8	inner ring
C	2	the two halves of the centre
D	8	between neighbouring A sensors
E	8	between neighbouring B sensors

Once started, the panel streams the state of all 34 sensors without being asked. Each report is a short fixed-length frame. Its seven payload bytes carry five sensor bits each:

$$7 \times 5 = 35 \text{ bits } \geq 34 \text{ sensors}$$

The game reads these frames on a background thread, independently of the display frame. The panel's own report rate is not stated anywhere in the game.

Start-up and sensitivity

At start-up the game resets each panel. It then writes a sensitivity value to every sensor, one at a time, and checks each echo. After three failed attempts the game raises a touch-panel error for that side.

The operator sets sensitivity as one of 11 levels, from −5 to +5, with 0 as the default. Each level maps to a raw value that the panel stores. The A sensors use their own scale, because they must react to a hand coming in from the edge. A lower raw value makes a sensor more sensitive. The game looks the raw value up by level and sensor group:

$$\text{raw value} = T_{\text{group}}(\text{level}), \qquad \text{level} \in \{-5, \dots, +5\}$$

Table 4. Raw sensitivity value T_{group} for each operator level.

Level	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5
A sensors	90	80	70	60	50	40	30	26	23	20	10
B, C, D, E sensors	70	60	50	40	30	20	15	10	5	1	1

The steps are not even. Below level 0 each step changes the raw value by 10. Above it the steps shrink, so the most sensitive settings change the panel only slightly.

2.5 Lighting

Each side has an LED board, part number 837-15070-04, on its own serial line at 115200 baud. The board has 11 lighting units, each with red, green and blue elements:

- one unit for each of the eight ring buttons;
- the cabinet body;
- the ring of light around the screen;
- the side panels.

The game sends at most one update every 33 ms. That caps the lighting at about 30 updates per second:

$$f_{\text{LED}} \le \frac{1000\ \text{ms}}{33\ \text{ms}} \approx 30.3\ \text{updates per second}$$

For each button LED the game keeps a fade: a start colour, an end colour and a duration. This lets it pulse the button that the player should press next. Before using a board, the game reads its board number and firmware checksum. Test mode shows both.

2.6 Cameras

The game uses up to three USB cameras. The two QR code readers, one per side, read codes that players show on their phones. The player camera takes the souvenir photos at the end of a credit. A camera that does not open within 5 s is reported as missing.

Table 5. Camera modes.

Camera	Resolution	Frame rate	Time per frame
QR code reader (1P, 2P)	640 × 480	5 fps	200 ms
Player camera	1280 × 960	30 fps	33.3 ms

The time per frame follows from the frame rate:

$$t_{\text{frame}} = \frac{1000\ \text{ms}}{\text{fps}}: \qquad \frac{1000}{5} = 200\ \text{ms}, \qquad \frac{1000}{30} \approx 33.3\ \text{ms}$$

The two modes differ widely in data rate. The player camera delivers about 24 times as many pixels per second as a QR reader:

$$\frac{1280 \times 960 \times 30}{640 \times 480 \times 5} = \frac{36{,}864{,}000}{1{,}536{,}000} = 24$$

A QR code needs only a sharp still image, so the low rate is enough. The game decodes these codes itself. It also generates the QR codes that it displays on screen.

2.7 Card reader, coins and credits

The cabinet service owns the Aime card reader. The reader accepts Aime cards, other FeliCa cards and the e-money cards used by the payment terminal. FeliCa is Sony's contactless card standard. The service passes the game a card identifier, and later the player's user id. The reader's LED can show red, green, blue and white. The service updates the reader's firmware at power-on when needed and shows the progress.

The cabinet service also counts coins, converts them to credits and keeps the books. A credit is the right to one play session. The machine holds at most 24 credits. Players can also buy credits through the e-money terminal, and the operator can set free play.

2.8 The game and the cabinet service

The game and the cabinet service run as two processes on the same PC. They talk through a local channel built from a named pipe and shared memory. The service starts first and stays up all the time. The game waits until the service is ready.

From then on the game asks the service for everything outside play itself:

- the button states and the coin counter;
- the credit count and the card that was just read;
- the machine's identity and region, and the version number shown on screen;
- the backup memory, which stores the operator's settings;
- the location of any additional data packages;
- the error to display when something fails.

In return, the game reports the start and end of each play, so the service can book the credit.

Game data comes as a base package plus optional add-on packages. The service mounts the add-ons, and the game reads them after the base package. An item in a later package replaces an item with the same id in an earlier one.

The network traffic of both programs, including ALL.Net authentication and the title server, is described in the network section.

2.9 Power-on

When the cabinet is switched on, the cabinet service starts first. The game then starts and shows a check screen on both monitors. Each check line ends in GOOD, BAD or N/A, or shows a blinking CHECK while it waits. The checks run in this order:

1. **System.** The game waits for the service. It then reads its settings, opens the inputs and LED boards, and switches all lighting off.
2. **Data.** The game loads its graphics and slide paths, then pauses 2 s.

3. **Card reader.** The game shows any firmware update in progress, then pauses 2 s.
4. **Network.** The service tests the network connection step by step. An IP address that does not arrive within 15 s counts as BAD.
5. **Title server.** The game waits until it reaches the game's server or gives up, then pauses 2 s.
6. **Devices.** A second screen checks both touch panels, both LED boards, the two QR readers, the player camera and the loading of all game data.
7. **In-store link.** If the shop links its cabinets, the game finds the other cabinets. It checks that only one claims to be the reference machine, copies that machine's settings, and joins the group for matching and shared demos.
8. **Attract mode.** After a final 2 s pause the game shows the warning and logo screen. It then plays demos and waits for a coin or a card.

A failing touch panel or camera produces a standard SEGA error code, which the service displays.

2.10 Test mode

The Test button opens the game test menu. The operator moves through it with the cabinet buttons.

Table 6. Entries of the game test menu.

Entry	What it shows or changes
Operation status	Coins, credits, plays by type, and power-on and play times
Input test	Buttons 1–8 and Select of each side, Service and Test
Output test	Each LED group of each side, the camera lamps and the coin blocker
Monitor test	A screen check page
Touch sensor test	The 34 sensors of each side, the sensitivity level and its adjustment
Camera test	Live picture and status of the three cameras
Game settings	See Table 7
Game system info	Software version, machine id, LED board details, add-on packages
Aime card reader test	Reader versions, a test read with its time, and an LED check
Closing-time settings	When the cabinet stops taking new credits: daily, per weekday or never
Network test	The network checks from power-on
Electronic payment info	Terminal id, brands, transaction history
VFD display test	A check of the cabinet's small fluorescent text display
ALL.Net billing log	Recent play history and closing information
ALL.Net download status	Current version and the progress of downloads
Clear backup data	Deletes operation data and unused credits
System test	Opens the cabinet service's own test mode

The game settings page holds six settings. The cabinet stores them in its backup memory.

Table 7. Settings on the game settings page.

Setting	Values	Effect
In-store matching	ON, OFF	Joins the shop's other cabinets for matching and shared demos
Reference machine	Reference, Follow	Chooses which cabinet sets the group's settings; forced to Reference when matching is off
Tournament mode	ON, OFF	Runs the cabinet in event mode; shown only on the reference machine
Tournament track count	1 to 4	Songs per credit in event mode
Continue	ON, OFF	Lets a player continue after the credit ends
Advertise volume	0, 25, 50, 75, 100 %	Volume of the attract-mode demos

The game menu has no price or free-play setting. Those belong to the cabinet service's system test mode, which all SEGA titles share.

3. Songs, charts and media

In brief. Each song bundles a description, up to six charts, one music track and a music video that plays behind the field. The 1.60 release ships 1,500 songs with 5,932 playable charts. The cabinet mixes the music live with a shared bank of 475 effect sounds and the voices of the system and 26 partners.

3.1 What a song carries

Figure 3 shows the parts of a song. The description holds the title, the artist, the genre, the tempo in beats per minute (BPM), the version that added the song and its unlock rule. The music-select screen sorts and filters by genre and by version. The stated tempo ranges from 55 to 339 BPM. It is only a label. The chart itself carries the tempo changes that drive play.

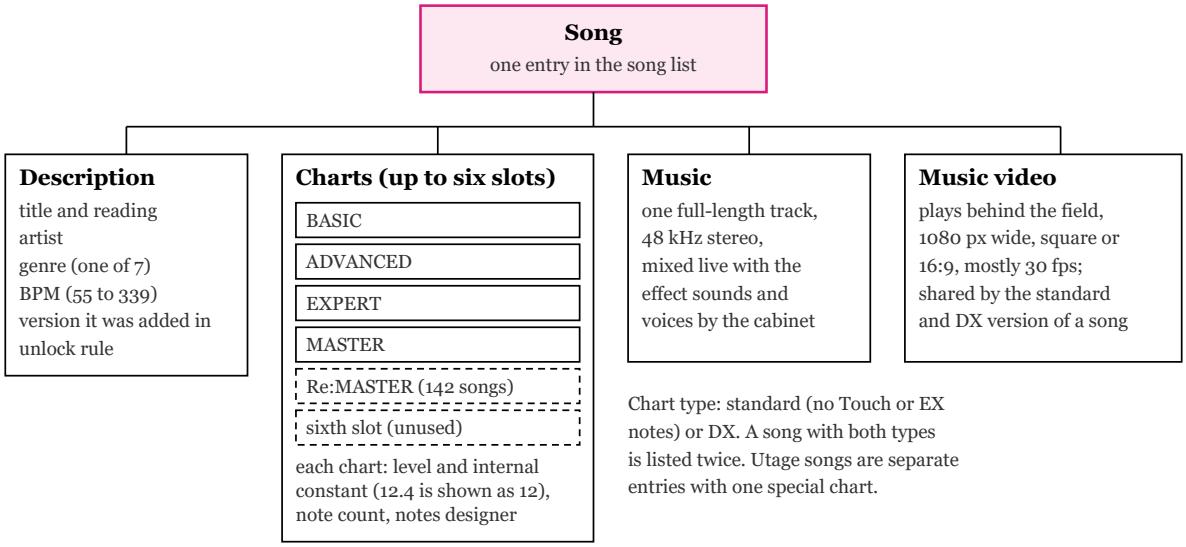


Figure 3. What one song consists of: its description, up to six chart slots with a level, internal constant and designer each, the music track and the music video.

Table 8 lists the seven genres.

Table 8. Songs per genre in release 1.60.

Genre	Songs
maimai	398
niconico & VOCALOID	369
GAME & VARIETY	254
Touhou Project	157
ONGEKI & CHUNITHM	141
POPS & ANIME	111
Utage hall	70

The unlock rule decides when a song appears. Most songs are open from the start. The others open through events, challenges or the KALEIDXSCOPE gates, which are a sequence of unlock stages. Table 9 gives the split.

Table 9. Unlock rules.

Rule	Songs
Open from the start	1,466
Locked until an event or reward	17
Opened through KALEIDXSCOPE	13
Challenge song	4

3.2 Charts and levels

Every song has six chart slots. The first four hold BASIC, ADVANCED, EXPERT and MASTER. The fifth holds Re:MASTER, an extra chart. No song in 1.60 uses the sixth slot. Each chart records a level, an internal constant, a notes designer and a note count. The internal constant is the level to one decimal place, such as 12.4. The player rating uses it. The highest constant among regular songs is 15.0. The game credits 186 named designers; the other charts show a dash.

The song list shows a whole level, with a "+" on the upper part of each level. Let c be the internal constant. The usual rule is

$$\text{shown level} = \begin{cases} \lfloor c \rfloor + & \text{if } c - \lfloor c \rfloor \geq 0.6 \text{ and } \lfloor c \rfloor \geq 7, \\ \lfloor c \rfloor & \text{otherwise.} \end{cases}$$

So 12.7 shows as "12+" and 12.4 as "12". Levels 1 to 6 have no plus step. The game stores the shown level separately, and a few charts with a constant of .6 or more still show without the plus.

A chart is either standard or DX. Standard charts use the note set of the older maimai cabinets. DX charts add Touch notes, which the player hits on the screen, and EX notes. Only 2 of the 2,426 standard charts contain Touch notes. 3,426 of the 3,434 DX charts do. A song with both versions appears as two entries that share music and video.

Utage charts are the "宴" party charts. Each utage song is its own entry with one special chart in the first slot. A single kanji labels its theme, such as 協 for a cooperative chart. The chart is played by one player, or by two players scoring together. An utage chart can also fix options while it plays. One cooperative chart turns off track skip, so nobody can end the song early.

3.3 How many songs and charts

Table 10 gives the counts for release 1.60.

Table 10. Songs, charts and media in release 1.60.

Item	Count
Songs	1,500
of which standard entries	583
of which DX entries	847
of which utage entries	70
Playable charts	5,932
BASIC, or the utage chart	1,500
ADVANCED	1,430
EXPERT	1,430
MASTER	1,430
Re:MASTER	142
Music videos	1,425
Shared effect sounds	475
System voice lines	150
Partner voice lines, 26 partners	2,404

3.4 The music video

A music video plays behind the round play field. 1,499 of the 1,500 songs point to one. There are fewer videos than songs because entries such as the standard and DX versions of a song share one. A song whose video is missing gets a short generic clip.

Almost all videos are 1,080 pixels wide. 931 are wide-screen at 1,080 by 608. 384 are square at 1,080 by 1,080 and fill the field exactly. The game centres a wide-screen video and letterboxes it. About 1,300 videos run at 30 frames per second, 76 at 24 and 39 at 60. 324 videos, mostly for older songs, are stills about a third of a second long. Videos do not loop, so the last frame stays on screen.

The video starts in the same frame as the music, but two video frames in. At a frame rate of f frames per second, the video leads the music by

$$\text{lead} = \frac{2}{f}\ \text{s},$$

which is 67 ms at 30 fps, 83 ms at 24 fps and 33 ms at 60 fps. The game does not correct drift after the start.

A black layer over the video keeps the notes readable. Let b be the opacity the player picks in the options: 0, 0.25, 0.4, 0.5 or 0.75, with 0.4 as the default. The layer starts fully black and clears to b over 500 ms after the start of play. Its opacity at time t in milliseconds is

$$\text{opacity}(t) = 1 - (1 - b)\,\min\!\left(1, \frac{t}{500}\right).$$

The game never plays sound from a video.

3.5 Sound design

Each song's music is one full-length stereo track at 48 kHz. The cabinet's sound engine mixes it live with effects and voices. Each group has its own volume category.

The shared bank gives each note type and judgement its own sound. Taps sound different for Critical Perfect, Perfect, Great and Good. Break notes have one sound for the higher judgements and one for the lower. EX notes, Slides, held Holds and Touch Holds have their own sounds too. So do FULL COMBO, ALL PERFECT and FULL SYNC. Table 11 lists the sound sets the player can choose in the options.

Table 11. Choosable sound sets.

Effect	Sets	Examples
Tap judgement	8	Default, Clap, Rain
Break	50	Default, Dog, Cat
EX note	7	Default, Synthesizer 1
Slide	50	Default, Dog, Cat

The play screen uses 212 of the 475 shared sounds. Table 12 shows where the rest go.

Table 12. Shared sounds outside the play screen.

Use	Sounds
Result screen	43
Map	37
KALEIDXSCOPE	33
Electronic-money payment	20
Entry	15
System	12
Track start	10
Short jingles: game start, game over, continue, map events	17
Background tracks for menus and results	13
Smaller groups on other screens	63

Voices live in banks of their own. The system voice has 150 lines. Each of the 26 partners, the companion characters a player can pick, has its own bank. Together they hold 2,404 lines.

4. Charts: notes, slides and touch sensors

In brief. A chart is a timed list of notes on a round play field. The player answers them with eight buttons around the screen and a touch panel of 34 sensors laid over it. Two things are worth remembering. The game measures chart time in bars of 384 ticks. It judges a slide by the order in which the finger crosses the sensors, not by the drawn path.

4.1 Songs and charts

Version 1.60 offers 1,500 songs and 5,932 playable charts. Of the songs, 583 have standard charts, 847 have DX charts and 70 are utage songs. Utage songs are special event charts with unusual rules or gimmicks.

The statistics below cover all 5,950 charts that the game carries. This set is slightly larger than the playable one for three reasons:

- The 20 two-player utage songs each hold two charts behind one playable slot.
- Two tutorial charts belong to no song.
- One song has four enabled charts that ship without chart data. The game can load such charts from data received over the network.

4.2 Chart time and tempo

The game counts chart time in bars and divides each bar into 384 ticks. A tick is the smallest step of the chart grid. A quarter note lasts 96 ticks, an eighth 48, a sixteenth 24, a triplet eighth 32 and a thirty-second 12.

Each note stores a single tick count from the start of the chart. The game converts it to milliseconds from the last tempo change at or before the note:

$$t_{\text{note}} = t_{\text{change}} + \frac{n_{\text{note}} - n_{\text{change}}}{384} \cdot 4 \cdot \frac{60\,000}{\text{BPM}}$$

where t_{note} is the note's time in milliseconds, n_{note} its tick count, n_{change} and t_{change} the tick count and time of the last tempo change, and BPM the tempo set by that change.

Example. A chart starts at 140 BPM and changes to 180 BPM at bar 16, which is tick 6,144. The change falls at $6\,144/384 \cdot 4 \cdot 60\,000/140 = 27\,428.6$ ms. A note at tick 6,720 lies 576 ticks, or six beats, later. At 180 BPM a beat lasts 333.3 ms, so the note falls at $27\,428.6 + 2\,000 = 29\,428.6$ ms.

The factor 4 means a bar always holds four beats for timing. The game reads the whole tempo list before it places any note, so a tempo change moves every later note. A chart can also change its meter. A meter change moves the bar lines and metronome clicks but not the notes.

Most charts keep one tempo:

- 5,394 charts use a single tempo.

- 556 charts change tempo at least once.
- 278 of those change tempo five times or more.
- 5,945 charts are in four-four time and 5 in six-eight time.

When the game loads a note, it records where the note falls within the beat. Table 13 shows how the 2.73 million notes are spread over those positions.

Table 13. Position of notes within the beat, all 5,950 charts.

Position	Notes	Share
On a quarter beat	1,983,063	72.7 %
On an eighth (not a quarter)	516,613	18.9 %
On a sixteenth (not an eighth)	151,999	5.6 %
On a twenty-fourth (triplet)	53,783	2.0 %
On a thirty-second	15,056	0.6 %
Elsewhere	6,137	0.2 %

The median chart ends at bar 89 and the longest at bar 191. The densest chart holds 2,222 notes.

4.3 Note approach time

The note speed option decides how long a note is visible before it must be hit. The game offers speeds from 1.00 to 10.00 in steps of 0.25, plus SONIC. Each speed has an internal value v, and the game derives one period D from it:

$$v = 100 \cdot s + 100, \qquad D = \frac{4 \cdot 60\,000}{v}\ \text{ms}, \qquad t_{\text{visible}} = 2D$$

where s is the displayed speed, v its internal value (5,000 for SONIC), D the length of one approach phase and t_{visible} the total time the note is on screen. During the first period the note grows in place at the center. During the second it travels out to the judgement ring. The tempo does not affect the approach time.

Example. The default speed is 4.50, so $v = 550$ and $D = 240\,000/550 = 436.4$ ms. The note appears 872.7 ms before its hit time. At speed 1.00 it appears 2,400 ms early, and at SONIC only 96 ms early.

Touch notes have their own speed option with a separate table of values; the default 4.50 maps to 350. How the game turns that value into time on screen was not traced.

4.4 Lanes and note types

The eight buttons sit on a ring around the display. They are numbered 1 to 8 clockwise, starting just right of the top. A button note travels outward from the center along one of eight lanes. The game judges it where it crosses the judgement ring, a circle of radius 480 on a display 1,080 units across.

The game knows five base kinds of note:

- **Tap**: a single press when the note reaches the ring.
- **Hold**: a press that starts at the ring and lasts for the hold's length.
- **Star**: a tap drawn as a star. It is the head of one or more slides that start from the same button at the same time.
- **Slide**: a path traced on the touch panel after the star. The star waits on the ring, then travels along the path, and the player follows it with a finger.
- **Touch** and **touch hold**: notes on a touch sensor instead of a lane. The player touches, or holds, the sensor when the closing markers meet.

Taps, holds, stars and slides come in variants:

- **Break**: worth more points.
- **EX**: any hit within nine frames, about 150 ms, either side of the target counts as Critical Perfect.
- **EX break**: both at once, for taps, holds and stars.

A chained slide consists of a first slide plus continuation pieces. Each piece starts where the previous one arrives, without a wait. Touch notes have no break or EX variant. Instead each carries a size, medium or large, and an optional **firework** flag that adds a burst effect when the note is hit. Figure 4 shows how each type looks on screen.

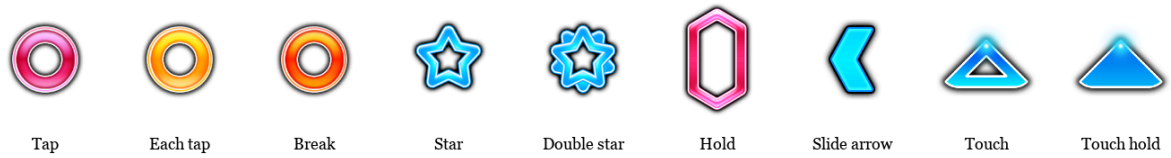


Figure 4. The note types as they appear on screen: tap, each tap, break, star, double star, hold, slide arrow, touch and touch hold.

Table 14 counts every type and variant across the shipped charts.

Table 14. Notes by type and variant, all 5,950 charts.

Kind	Normal	Break	EX	EX break
Tap	1,699,670	86,462	100,874	10,686
Star (slide head)	217,902	10,717	11,693	1,597
Hold	202,351	2,255	12,210	1,584
Slide	235,839	6,282	0	0
Touch	116,043	–	–	–
Touch hold	3,799	–	–	–

Another 6,652 continuation pieces belong to chained slides. EX slides exist in the game, but no shipped chart uses them. The charts hold 241,909 star heads for 242,121 slides, because one star can launch several slides. A star with more than one slide is drawn as a double star.

Table 15 summarizes the 119,842 touch notes and touch holds.

Table 15. Touch notes by size and effect, all 5,950 charts.

Property	Count
Medium size	87,148
Large size	32,605
No size given (shown as medium)	89
With firework flag	8,791

In practice a chart gives all its touch notes the same size. The game also offers four mirror modes: none, left-right, top-bottom and a 180-degree turn. The two true mirrors swap every left-handed slide shape with its right-handed twin. The 180-degree turn keeps them.

4.5 Each notes

Each notes are notes that must be hit together. The game draws them in yellow and joins them with a bar. Chart authors do not mark them; the game works them out with this rule:

$$\text{each}(a, b) \iff t_a = t_b \ \wedge\ \text{slide}(a) = \text{slide}(b) \ \wedge\ \neg\,\text{piece}(a) \ \wedge\ \neg\,\text{piece}(b)$$

where t_a and t_b are the two notes' tick counts, $\text{slide}(x)$ is true when note x is a slide, and $\text{piece}(x)$ is true when x is a continuation piece of a chained slide.

Example. A tap on button 2 and a star on button 6 share a tick. Neither is a slide, so they form an each pair. The star's slide starts at that same tick, but it pairs only with another slide at that tick, never with the tap.

Simultaneous touch notes also form chains when their sensors are neighbors. A B sensor, for instance, links to the B sensors one lane either side, the E sensors beside it, the A sensor in front of it and the center. A chain shares one hit effect.

4.6 Hold and slide lengths

Holds and slides have lengths in ticks. A slide has two: the wait before the star leaves the ring, and its travel time. The wait is one beat in 241,318 slides and zero in 6,714, nearly all continuation pieces. The star therefore rests for exactly one beat at any tempo. Table 16 lists the most common lengths.

Table 16. Most common hold lengths and slide travel times, all 5,950 charts.

Rank	Hold length	Slide travel time
1	1 beat	½ beat (141,439 slides)
2	2 beats	1 beat
3	½ beat	2 beats
4	¾ beat	3 beats
5	1 bar	1½ beats

4.7 Slide shapes

The game has thirteen slide shapes. It defines each shape for a start at button 1 and an end offset. The end offset says which button the slide ends on:

$$o = (e - b) \bmod 8 \quad \text{for left-handed shapes,} \qquad o = (b - e) \bmod 8 \quad \text{for right-handed shapes}$$

where b is the start button, e the end button and o the end offset. Counting runs clockwise for left-handed shapes and counter-clockwise for the five right-handed ones.

Example. A clockwise arc from button 1 to button 3 is a right-handed shape. Its offset is $(1 - 3) \bmod 8 = 6$. The counter-clockwise arc from button 1 to button 7 is left-handed, with offset $(7 - 1) \bmod 8 = 6$. The two are mirror images, so the game stores one and reflects it for the other.

Not every offset exists for every shape. Table 17 lists the shapes, the offsets the game can draw and how often the charts use each one.

Table 17. The thirteen slide shapes, all 5,950 charts (variants and continuation pieces included).

Shape	Course	End offsets	Slides
Straight	a straight line across the field	2–6	135,978
Arc, left and right	along the outer ring	0–7 (0 = full circle)	33,824 and 34,475
U-curve, left and right	a U through the inner ring	0–7	6,996 and 7,049
Thunder, left and right	a zig-zag through the center	4 only	1,527 and 1,316
V	into the center and out again	1–3, 5–7	7,082
Loop, left and right	a loop around the center	0–7	5,640 and 5,533
L-shape, left and right	to a button two lanes away, then across	1–4	3,126 and 3,197
Fan	three straight arms to neighboring buttons	3–5	3,030

4.8 Slide geometry

The game draws each path from a vector outline on a field 1,080 units square. The buttons sit on a circle of radius $R = 480$ around the center. The outlines are made of cubic Bézier curves. The game samples each curve 32,768 times and keeps one point every 47.112 units of path length. It rotates the base path in 45-degree steps for the other start buttons. It builds each right-handed shape by

reflecting the left-handed one about the axis through button 1. Figure 5 shows one example of each shape.

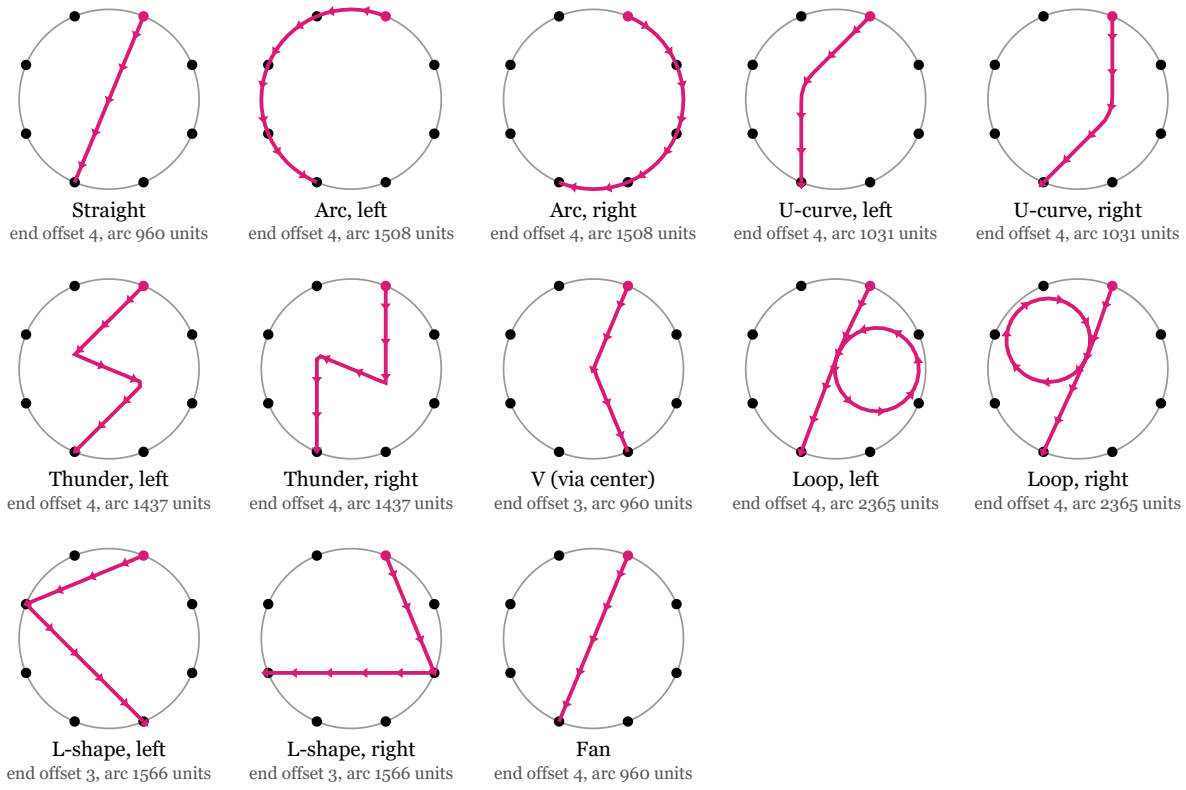


Figure 5. The thirteen slide shapes for a start at button 1 (accent dot), one end offset each, with their path lengths in field units; the fan shows its middle arm.

The path lengths follow from the ring geometry:

$$L_{\text{straight}} = 2R\sin\left(\frac{\pi k}{8}\right), \qquad L_{\text{arc}} = R \cdot \frac{\pi}{4} \cdot m, \qquad N \approx \left\lfloor \frac{L}{47.112} \right\rfloor + 2$$

where k is the number of lanes between the ends of a straight slide, m the number of lanes an arc passes, L the path length in field units and N the number of points the game keeps along it, end point included.

Example. A straight slide from button 1 to button 5 spans $k = 4$ lanes. Its length is $2 \cdot 480 \cdot \sin(\pi/2) = 960$ units, the ring's diameter. An arc over four lanes is $480 \cdot \pi = 1\,508$ units long, half the circumference. The game keeps $\lfloor 1\,508 / 47.112 \rfloor + 2 = 34$ points along it. The arrow chevrons sit on these points.

4.9 Tracing a slide through the sensors

The game does not judge a slide by its drawn path. It uses an ordered list of sensor steps for every shape, start and end offset. Each step names one or more sensors that count as the same step. Each step also has a push distance and a release distance. The game judges a slide as follows:

1. The player touches the steps in order. A step is done when its sensor is touched and released. The last step needs only a touch.
2. While one step is pending, the game also accepts a touch on the next step, so a fast finger may skip one sensor.
3. On lists of three steps or fewer, the final step counts only after the step before it. The L-shapes have their own skip rule.
4. When the last step is touched, the game adds an allowance for the remaining travel and compares the result with the star's arrival time.
5. The game shows the judgement once the allowance has passed.

The allowance is the travel time scaled by the last step's share of all distances:

$$w = T \cdot \frac{p_{\text{last}}}{\sum_{i} \left(p_i + r_i\right)}$$

where w is the allowance in milliseconds, T the star's travel time, p_i and r_i the push and release distances of step i, and p_{last} the push distance of the last step.

Example. A straight slide from button 1 to button 5 has five steps: A1, B1, C, B5 and A5. Their distances sum to 960.4 units, and the last push distance is 156.4. With a travel time of half a beat at 150 BPM, $T = 200$ ms, the allowance is $200 \cdot 156.4 / 960.4 = 32.6$ ms.

Table 18 lists the sensor steps for the shapes in Figure 5. For other start buttons the game rotates the sensor names by the start position. The center counts as one sensor, whichever half is touched.

Table 18. Sensor steps for a slide starting at button 1.

Shape	End offset	Sensors in order
Straight	4	A1, B1, C, B5, A5
Arc, left	4	A1, A8, A7, A6, A5
Arc, right	4	A1, A2, A3, A4, A5
U-curve, left	4	A1, B8, B7, B6, A5
U-curve, right	4	A1, B2, B3, B4, A5
Thunder, left	4	A1, B8, B7, C, B3, B4, A5
Thunder, right	4	A1, B2, B3, C, B7, B6, A5
V	3	A1, B1, C, B4, A4
Loop, left	4	A1, B1, C, B4, A3, A2, B1, C, B5, A5
Loop, right	4	A1, B1, C, B6, A7, A8, B1, C, B5, A5
L-shape, left	3	A1, A8 or B8, A7, B6, B5, A4
L-shape, right	3	A1, A2 or B2, A3, B4, B5, A6
Fan	4	A1, B1, C, B5, A5

4.10 The touch panel

The touch panel has 34 sensors in five groups. The panel reports only which sensors are touched, not where the fingers are. A chart places a touch note by group and position; a note on C ignores the position. Table 19 describes the groups, and Figure 6 draws them in their true shapes.

Table 19. The five sensor groups, sizes in field units, and their use in all 5,950 charts.

Group	Sensors	Shape and place	Touch notes
A	A1–A8	sectors in front of the buttons, radius 326 to 537	6,863
B	B1–B8	inner sectors at the same angles, radius 151 to 285	54,021
C	C1, C2	the center, split into right and left halves of 99 × 208	13,513
D	D1–D8	wedges between the A sensors, radius 364 to 537, D1 at the top	4,791
E	E1–E8	squares of side 100, turned 45°, centered at radius 310 between the B sensors	40,654

Buttons 1-8 around the judgement ring (r = 480); 34 touch sensors; display edge r = 540

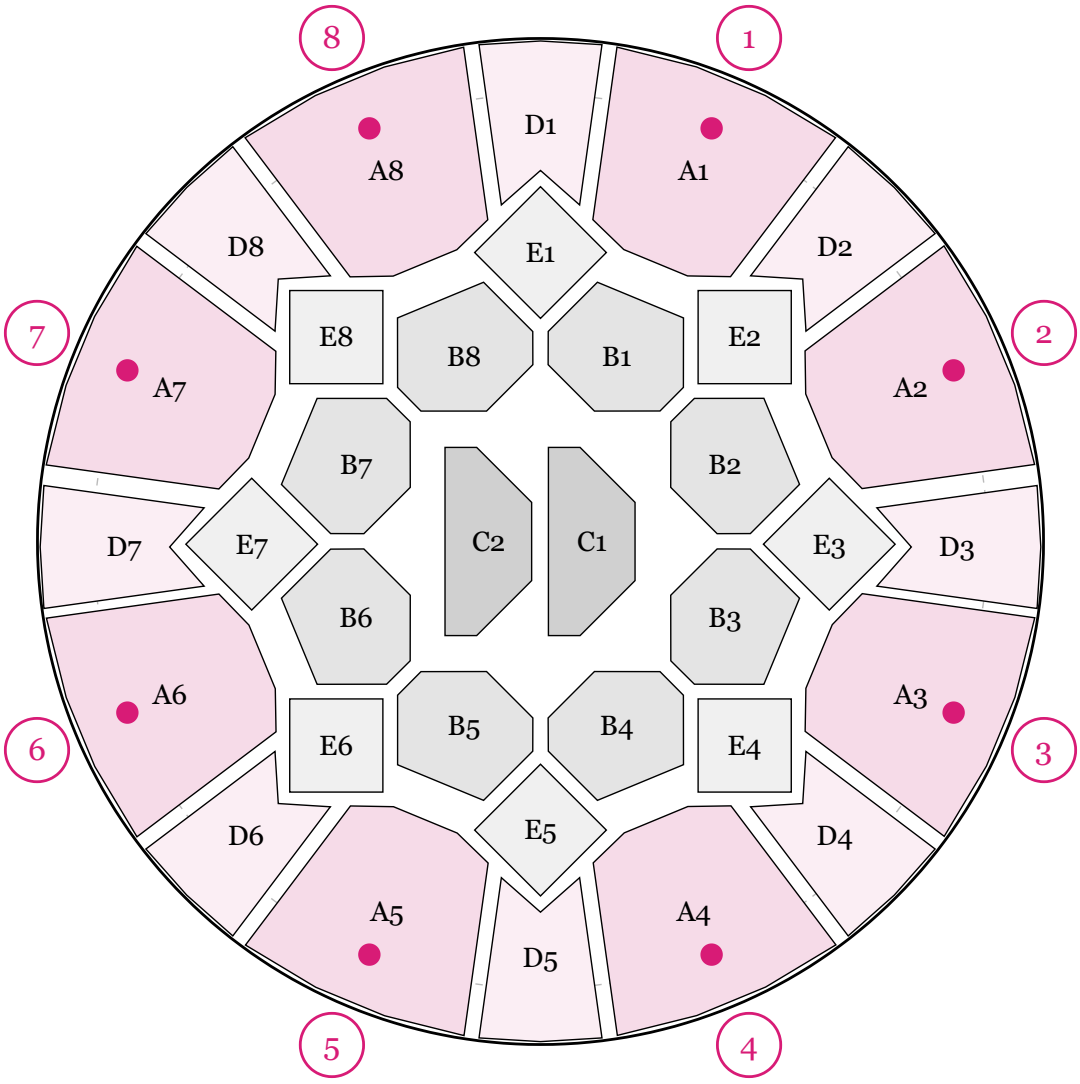


Figure 6. The eight buttons, numbered, around the judgement ring of radius 480, and the 34 touch sensors in their true shapes; the display edge is at radius 540.

4.11 Difficulties, levels and note counts

Every song has six chart slots: Basic, Advanced, Expert, Master, Re:Master and a sixth slot reserved for utage. Each slot holds a level, a decimal level, the note designer's name, the note count and an on-off switch. Table 20 shows how the songs fill the slots.

Table 20. Playable charts per slot.

Slot	Songs with a chart
Basic (or the single utage chart)	1,500
Advanced	1,430
Expert	1,430
Master	1,430
Re:Master	142
Sixth slot	0

Twenty utage songs are written for two players at once. Each holds a left-player chart and a right-player chart behind its first slot. 61 charts are marked as utage charts. Among them is a 274 BPM chart with 53 touch notes.

The game sorts levels on a ladder: 1 to 7, then 7+, 8, 8+ and so on up to 15+. A decimal level of .6 or more shows as a plus level. This threshold matches the shipped data but was not confirmed in the game logic. True Love Song, for example, offers Basic 5.0, Advanced 7.2, Expert 10.2 and Master 12.4 with 96, 122, 177 and 302 notes.

The note count is the number of taps, stars, holds, slides, touch notes and touch holds. Continuation pieces do not count. The note count sets the ceiling of the DX score, a secondary score that rewards precise timing:

$$S_{\text{DX,max}} = 3 \cdot N_{\text{notes}}$$

where N_{notes} is the chart's note count. *Example*. The Master chart of True Love Song has 302 notes, so its DX score ceiling is 906. The game computes the chart's theoretical maximum score from the notes themselves when it loads the chart.

5. Judgement, scoring, ranks and rating

In brief. The game turns every press, slide and touch into one of five judgements, and fixed tables turn those judgements into points, achievement, DX score, medals, ranks and rating. Two numbers matter most: a perfect play scores 101.0000 %, and the rating of a chart grows with its level and achievement up to a cap at 100.5000 %.

5.1 Judgements and timing windows

The game uses five judgements: Critical Perfect, Perfect, Great, Good and Miss. A Critical Perfect is the most exact hit. Internally the game splits the Perfect band in two and the Great band in three, and it marks every non-Critical hit as fast or late. That gives fifteen timing bands per note.

A judgement window is the span of time around a note in which a hit earns a given judgement. The game stores each window edge in frames of 16.666666 ms. It then shifts every edge 50 ms later, so all windows centre 50 ms after the chart time of the note. The player's timing option shifts them further.

$$t_{\text{edge}} = 16.666666 \times (f + o) + 50$$

where t_{edge} is the edge in ms after the chart time, f is the edge in frames from Table 21, and o is the timing option in frames (−2.0 to +2.0 in steps of 0.1, default 0).

Example. The late edge of a tap's Good band is 9 frames. With the default option it lies at $16.666666 \times 9 + 50 = 200$ ms after the chart time, which is +150 ms from the centre.

Table 21. Window edges per note type, in frames from the centre (fast miss edge first, late miss edge last).

note type	edges in frames	what the edges mean in ms from the centre
Tap, star, hold head	−9 −6 −5 −4 −3 −2 −1 0 1 2 3 4 5 6 9	Critical ±16.7, Perfect ±50, Great ±100, Good ±150
Break (tap, star, hold head)	same as Tap	same as Tap; each one-frame band pays different points
Touch, touch-hold head	−9 ×7, 0, 9 10.5 12 13 14 15 18	Critical −150 to +150; late Perfect to +200, late Great to +250, late Good to +300
EX note	−9 ×7, 0, 9 ×7	Critical ±150, otherwise Miss
Slide end	−36 −26 −22 −18 −14 −14 −14 0 14 14 14 16 22 26 36	Critical ±233.3, no Perfect band, Great to −433.3 and +433.3, Good ±600

Figure 7 draws the tap and break windows to scale, with the points each band pays. The fast edge of each band is exclusive and the late edge inclusive.

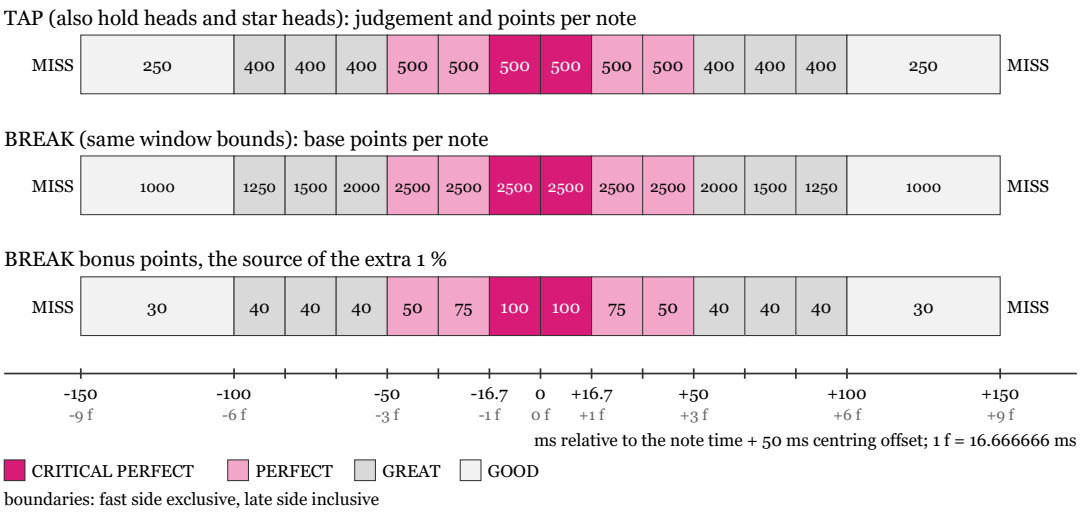


Figure 7. Judgement windows of a tap and the points each band is worth; the break rows show how the 2500-point break and its bonus fall off one frame at a time.

The game judges a press as follows.

1. A tap becomes judgeable at its fast miss edge, 150 ms before its centre.
2. The first press of its button after that moment is judged.
3. Only the earliest unfinished tap or hold on a lane can take a press, so a press never skips to a later note.
4. If no press comes before the late miss edge, the note is a Miss.

The FAST or LATE number on screen is the offset from the centre, not from the chart time.

5.2 Holds and touch-holds

The game judges a hold twice. The head is judged like a tap. For the body, the game watches the button from 100 ms after the head time until 200 ms before the tail time. For a touch-hold the watch starts 250 ms after the head. Every frame without a press adds to a release total. At the tail the game computes the released share:

$$p = \min\left(100,\ \left\lceil 100 \times \frac{r}{L - h - 200} \right\rceil\right)$$

where p is the released share in percent, r is the total release time in ms, L is the hold length in ms, and h is 100 ms (250 ms for a touch-hold).

Example. A 2000 ms hold released for 300 ms gives $\lceil 100 \times 300/1700 \rceil = 18$ %. A head judged Critical becomes a late Perfect (Table 22).

Table 22. Final judgement of a hold by released share and head judgement.

released share	head Critical	head Perfect	head Great	head Good	head Miss
0 %	Critical	Perfect	Great	Good	Miss
1–33 %	late Perfect	Perfect	Great	Good	Miss
34–67 %	late Great	Great	Great	Good	Miss
68–95 %	late Good	Good	Good	Good	Miss
96–100 %, or never held	late Good	Good	Good	Good	Miss

Two shortcuts apply. A hold no longer than 300 ms (450 ms for a touch-hold) keeps its head judgement. The fast or late side of the head survives into the final judgement. The game scores a touch-hold as a hold, not as a touch.

5.3 Slides

A slide is a star that launches from its tap and travels a path to an end lane. The game tracks the slide with hit areas, which are groups of touch sensors along the path. The player enters an area by pressing any of its sensors. The game allows skipping one area on slides with more than three areas, and on shorter slides as long as the skip does not land on the last area.

The judgement happens when the player reaches the last area. The game times it against the moment the star itself enters that area. It also widens the Critical band by a quarter of the time the star spends crossing the last area:

$$w = T \times \frac{d_{\text{last}}}{d_{\text{total}}}, \qquad \text{Critical if } -183.3 - \frac{w}{4} \le \Delta \le 283.3 + \frac{w}{4}$$

where w is the last-area crossing time in ms, T is the star's travel time, d_{last} and d_{total} are the length of the last area and of the whole path, and Δ is the player's arrival minus the star's entry into the last area, in ms.

Example. A star travels for 1000 ms and the last area covers 20 % of the path. Then $w = 200$ ms and the Critical band runs from −233.3 ms to +333.3 ms around the star's entry.

Outside that band the slide row of Table 21 applies. Two softer rules follow:

- a slide finished too early counts as a fast Good, not a Miss;
- a slide that times out with every area except the last traced counts as a late Good.

So a slide misses only when its last area is never reached.

5.4 Points and achievement

Each judgement pays fixed points by note type (Table 23).

Table 23. Points per judgement and note type.

note type	Critical	Perfect	Great	Great, 2nd and 3rd band	Good	Miss
Tap, Touch	500	500	400	400	250	0
Hold, touch-hold	1000	1000	800	800	500	0
Slide	1500	1500	1200	1200	750	0
Break, base	2500	2500	2000	1500, 1250	1000	0
Break, bonus	100	75, 50	40	40	30	0

Achievement is the percentage score shown during and after play. The game computes it from the base and bonus points:

$$A = 100 \times \frac{B}{B_{\max}} + \frac{K}{K_{\max}}$$

where A is the achievement in percent, B is the base points earned, $B_{\max}$ is the sum of the Critical base points of all notes, K is the break bonus earned, and $K_{\max}$ is 100 times the number of breaks. On a chart without breaks the second term is dropped.

Example. A chart has 400 taps, 100 holds, 50 slides and 20 breaks, so $B_{\max} = 425\,000$ and $K_{\max} = 2000$. One Great on a tap and every other note Critical gives $A = 100 \times 424\,900 / 425\,000 + 1 = 100.9765\ \%$.

The base term tops out at 100 % and the bonus term at 1 %, so a perfect play scores 101.0000 %. The game keeps achievement as a whole number of ten-thousandths of a percent, so 101.0000 % is stored as 1 010 000. Two-player utage charts double every ceiling and threshold.

5.5 DX score

The DX score counts exactness. Each note earns 3 for Critical, 2 for Perfect, 1 for Great and 0 otherwise. The result screen awards DX stars by the share of the maximum:

$$D_{\max} = 3N, \qquad q = \left\lfloor 100 \times \frac{D}{D_{\max}} \right\rfloor$$

where D is the DX score, N is the number of notes, and q is the whole percentage compared with Table 24.

Example. A 570-note chart has $D_{\max} = 1710$. A DX score of 1650 gives $q = 96$, which earns four stars.

Table 24. DX stars.

stars	1	2	3	4	5
share of maximum ≥	85 %	90 %	93 %	95 %	97 %

5.6 Combo, FAST and LATE

The combo counter rises on every judgement except Miss and resets to zero on a Miss. At the end of a track the game awards a combo medal (Table 25).

Table 25. Combo medals. All require no Miss.

medal	condition
ALL PERFECT+	achievement exactly 101.0000 % (100.0000 % on a chart without breaks)
ALL PERFECT	every note Perfect or Critical
FULL COMBO+	every note Great or better
FULL COMBO	no Miss

The FAST and LATE counters follow the judgement-display option. With the default display the game counts only fast and late Greats and Goods. Displays that show Perfect timing also count Perfects.

5.7 Sync

When two to four players play the same song on one cabinet, a shared chain counter sums their combos and resets when anyone misses. At the end the game compares each pair of players whose medals allow it and keeps the best sync medal (Table 26).

Table 26. Sync medals.

medal	rule	sync points
FULL SYNC DX+	both players have ALL PERFECT or better	12
FULL SYNC DX	both have FULL COMBO+ or better, not both ALL PERFECT	10
FULL SYNC+	one of the two has only FULL COMBO	5
FULL SYNC	the player is on a higher difficulty than the partner	2
SYNC PLAY	no sync medal, two or more players on the song	1

5.8 Life

Life is a counter that only Perfect Challenge, the course modes and KALEIDXSCOPE use. Each frame the game subtracts the damage of new judgements. At zero the player fails.

$$L' = \max\left(0,\; L - \left(n_P\, d_P + n_G\, d_G + n_g\, d_g + n_M\, d_M\right)\right)$$

where L and L' are life before and after, n_P, n_G, n_g, n_M are the new Perfects, Greats, Goods and Misses, and d_P, d_G, d_g, d_M are their damage values from Table 27.

Table 27. Damage per judgement by mode.

mode	starting life	Perfect	Great	Good	Miss
Perfect Challenge	set per chart	0	1	1	1
KALEIDXSCOPE gate	1 to 100, by phase	0	1	1	1
KALEIDXSCOPE EX life	100, 200, 300, …	1	10	20	50
course	set per course (Table 28)	set per course			

In a course, life carries over the four tracks. After each track the game adds the course's recovery, never above the starting life:

$$L_{\text{next}} = \min(L + R,\ L_0)$$

where R is the recovery per track and L_0 the starting life.

Example. In a 四段 course ($L_0 = 700$, $R = 50$, Great 2, Good 2, Miss 5), a track with 10 Greats, 4 Goods and 3 Misses costs $20 + 8 + 15 = 43$. Life falls from 700 to 657, then recovers to 700.

The game holds 294 courses of four songs each, in three modes: dan (100 courses), true dan (117) and random dan by difficulty (77).

Table 28. Course life settings.

mode	life	recovery	damage Perfect / Great / Good / Miss	courses
dan, 初段 to 二段	350	20	0 / 0 / 2 / 5	20
dan, 三段	600	50	0 / 1 / 2 / 5	10
dan, 四段 to 八段	700	20 or 50	0 / 2 / 2 / 5	50
dan, 九段 and 十段	800 to 900	20 to 50	0 / 2 / 2 / 5	20
true dan	50	10 (5 for 真皆伝)	0 / 2 / 3 / 5	110
true dan, 裏皆伝	10	0	0 / 1 / 3 / 10	7
random dan	700 down to 100	50 down to 10	0 / 2 / 2 or 3 / 5	77

5.9 Clear ranks

The rank is the highest rank whose threshold the achievement reaches:

$$\text{rank}(A) = \max\{\, r : A \geq \theta_r \,\}$$

where θ_r is the threshold of rank r from Table 29.

Example. 99.7321 % reaches SS+ (99.5) but not SSS (100), so the rank is SS+.

Table 29. Clear-rank thresholds.

rank	D	C	B	BB	BBB	A	AA	AAA	S	S+	SS	SS+	SSS	SSS+
achievement ≥ (%)	0	50	60	70	75	80	90	94	97	98	99	99.5	100	100.5

A track counts as cleared from rank A. Figure 8 lists the same thresholds next to the rating coefficient at each line.

Clear ranks and the rating coefficient reached at each rank line

rank	threshold (x 10000)	achievement %	coefficient k	note
D	0	0.0000	0	below 50 %
C	500000	50.0000	80	
B	600000	60.0000	96	
BB	700000	70.0000	112	
BBB	750000	75.0000	120	
A	800000	80.0000	136	cleared from rank A
AA	900000	90.0000	152	
AAA	940000	94.0000	168	
S	970000	97.0000	200	coefficient jumps 176 to 200
S+	980000	98.0000	203	
SS	990000	99.0000	208	
SS+	995000	99.5000	211	
SSS	1000000	100.0000	216	
SSS+	1005000	100.5000	224	maximum 101.0000 %

A play takes the highest rank whose threshold it reaches. Two-player utage charts double every threshold.

Figure 8. Clear-rank thresholds, with the rating coefficient that applies at each rank line.

5.10 DX rating of one chart

The DX rating measures a player's skill. The cabinet computes it from the player's best scores. One chart is worth:

$$R_{\text{chart}} = \left\lfloor \frac{c \times \min(a,\ 1\,005\,000) \times k(a)}{10^8} \right\rfloor + b$$

where c is ten times the chart constant (143 for 14.3), a is the achievement in ten-thousandths of a percent, $k(a)$ is the coefficient from Table 30, and b is 1 for ALL PERFECT or ALL PERFECT+ and 0 otherwise.

Example. A 14.3 chart at 100.5000 % gives $\lfloor 143 \times 1\,005\,000 \times 224 / 10^8 \rfloor = 321$, or 322 with ALL PERFECT. A 13.0 chart at 100.0000 % gives 280.

Table 30. Rating coefficient by achievement. The coefficient is that of the highest row reached.

achievement ≥ (%)	0	10	20	30	40	50	60	70	75	79.9999	80	90
k	0	16	32	48	64	80	96	112	120	128	136	152

achievement ≥ (%)	94	96.9999	97	98	98.9999	99	99.5	99.9999	100	100.4999	100.5
k	168	176	200	203	206	208	211	214	216	222	224

Figure 9 plots the coefficient as a step function. The rows one ten-thousandth below a rank line pay a play just short of the line an in-between value. The cap means 100.6 % and 101 % are worth the same as 100.5 %.

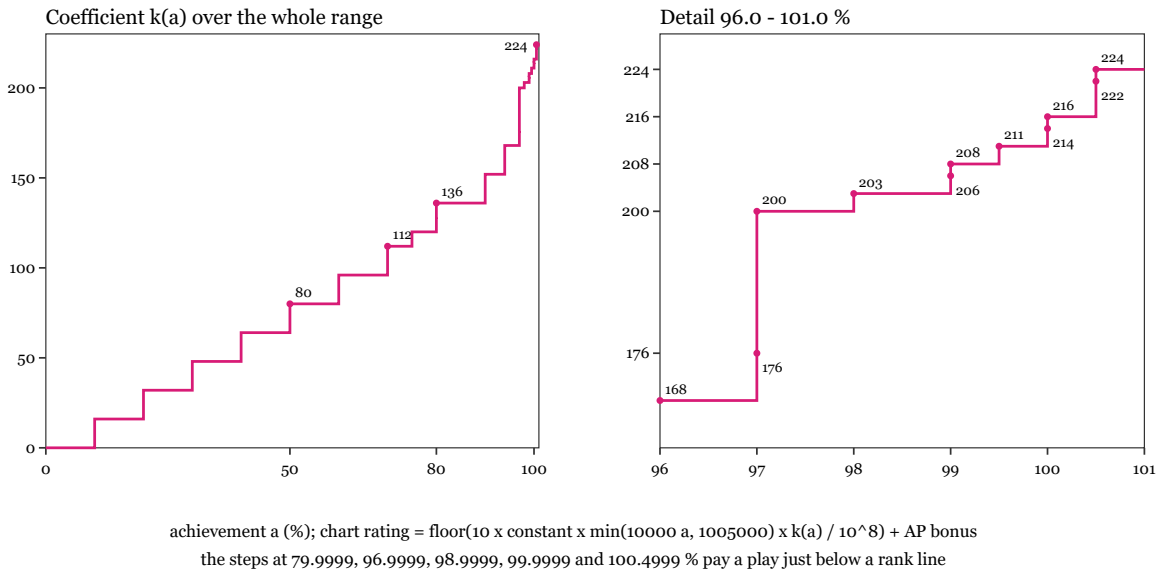


Figure 9. The rating coefficient as a step function of achievement; the right panel enlarges 96 to 101 %, where the steps are dense.

5.11 Total rating

The player's rating is the sum of two lists of best charts:

$$R_{\text{total}} = \sum_{i=1}^{35} R_i^{\text{old}} + \sum_{j=1}^{15} R_j^{\text{new}}$$

where R_i^{old} are the 35 best chart ratings among songs from earlier versions and R_j^{new} the 15 best among songs from the previous and the current version (1.55 and 1.60).

Example. Old-list entries averaging 280 and new-list entries averaging 300 give $35 \times 280 + 15 \times 300 = 14\,300$, a Gold plate.

A new score replaces the stored entry for its chart only if it is worth more. In the 1.60 data, 123 of 1500 songs fall in the new group. The game also prepares two lists of up to 20 charts that could still raise the rating. The rating plate colour follows the total (Table 31).

Table 31. Rating plate colours.

colour	White	Blue	Green	Yellow	Red	Purple	Bronze	Silver	Gold	Platinum	Rainbow
rating ≥	0	1000	2000	4000	7000	10000	12000	13000	14000	14500	15000

5.12 Class ladder

Ghost and friend battles use a ladder of 26 classes: B5 to B1, A5 to A1, S5 to S1, SS5 to SS1, SSS5 to SSS1, and LEGEND. After each battle the game updates the player's class points:

$$P' = P + \delta, \qquad \text{promote if } P' \geq \beta, \quad \text{demote if } P' < 0$$

where P is the current class points, δ is the win or loss value from Table 32 for the opponent's class, and β is the class border.

Example. An S3 player ($\beta = 30$) with 28 points beats an S2 opponent and gains 4, reaching the border. The points then stay at the border until the player beats the class boss. That win moves the player up to S2 with 10 points.

Table 32. Class ladder settings. Win and loss values are against an opponent of the same, a higher or a lower class.

classes	border	start after promotion / demotion	win same / higher / lower	loss same / higher / lower
B5–B1	10	0 / 5	5 / 5 / 5	0 / 0 / 0
A5	10	0 / 5	4 / 4 / 3	0 / 0 / 0
A4–A1	20	10 / 10	4 / 4 / 3	−1 / −1 / −1
S5–S1	30	10 / 15	3 / 4 / 3	−1 / −1 / −2
SS5–SS1	50	10 / 25	3 / 3 / 2	−2 / −1 / −3
SSS5 to SSS1	60, 70, 80, 90, 100	10 / 30 to 45	2 or 3 / 3 / 1 or 2	−2 / −1 / −3
LEGEND	9999	0 / 0	1 / 1 / 1	0 / 0 / −1

A win against an opponent three or more classes lower earns nothing. Demotion never goes below B5.

5.13 The play record

At the end of each track the cabinet sends a play record to the title server. At the end of the credit it sends the whole updated profile. The play record holds:

- who played, where and when, and the track number;
- the song, the difficulty and any co-players;
- the achievement, DX score, rank, maximum combo and chain;
- the five judgement counts for each note type, and the FAST and LATE counts;

- the combo and sync medals, the clear flag and any new records;
- the rating before and after the play.

The profile update carries the best record of every chart, both rating lists, the class ladder state, course and KALEIDXSCOPE progress, and the credit's play records.

6. A credit, screen by screen

In brief. A credit is one paid session, normally three songs called tracks, played as a fixed chain of timed screens. This section follows that chain, then covers options, collections and the reward systems between credits. The key formula is the area distance, which multiplies the touring team, the play bonuses, the track count and the ticket.

6.1 The screen sequence

Figure 10 shows the chain for one credit. Each box gives the screen's countdown or length. Each arrow gives the event that moves the game on. Start-up checks run once at power-on and are not shown.

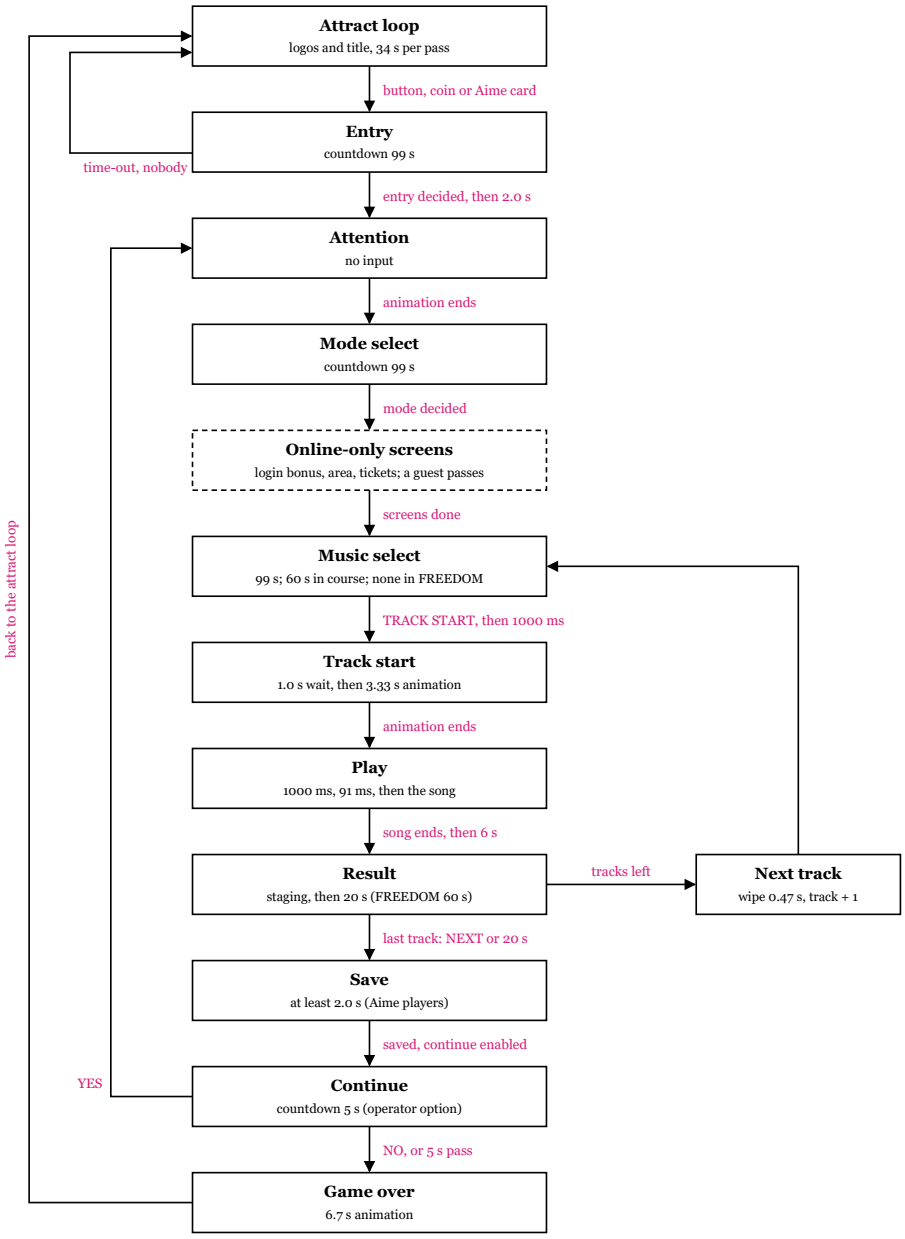


Figure 10. The screens of one credit in order, with each screen's countdown or duration and the event that moves the game on; the dashed box groups the online-only screens a guest passes through.

Table 33 lists the countdown of every screen that has one. The countdown is the timer shown on screen. When it reaches zero, the game makes the choice for the player.

Table 33. Countdowns of the screens in one credit.

Screen	Countdown	What happens at zero
Entry	99 s	back to the attract loop if nobody entered
Mode select	99 s	the centred mode is chosen
Login bonus	60 s	the screen closes
Area select	30 s	the current area is kept
Tour-member select	60 s	the current team is kept
Ticket select	99 s	no ticket is used
Tutorial offer	10 s	the tutorial is skipped
Music select	99 s (60 s in course and KALEIDXSCOPE, none in FREEDOM)	the selected song starts
Result	20 s, hidden (60 s in FREEDOM)	the game moves on
Continue	5 s	NO is chosen

The countdown digits are white from 10 s up and red below 10 s. A warning sound plays at 30, 20 and 10 s, and a tick plays each second from 9 down.

6.2 Attract loop

The attract loop is the demo the cabinet plays while idle. One pass is a 1.0 s pause, the SEGA and ALL.Net logos (14.0 s) and the title (19.0 s), so 34 s in all. With an online popularity ranking, a 60 s demo play of the top song follows.

A credit starts on a button press with enough credit, a coin, or an Aime card. The Aime card is SEGA's player ID card.

6.3 Entry

The entry screen decides who plays. It starts the entry music, which runs on until the first category screen. Button 4 is GUEST PLAY and button 7 is ACCESS CODE. Touching an Aime card logs the player in. A guest plays without saving any results.

A guest presses GUEST, then YES in a warning window. The other side may still join. After an "entry complete" line and 2.0 s, the game moves on. Other buttons take 30 s off the countdown.

6.4 Attention, information and mode select

The attention screen asks players not to hit or rub the panel hard. It takes no input. Meanwhile the game loads each player's partner voices (see "Partners and voices"). If both players entered, it grants a fourth track. An information screen with event news follows for returning Aime players only.

Mode select shows four modes on a turning carousel:

- **NORMAL:** three tracks of the player's choice.
- **DAN COURSE:** a fixed set of songs played against a life gauge.
- **FREEDOM:** free song choice for a fixed time.
- **KALEIDXSCOPE:** a gate challenge with set songs.

Buttons 3 and 6 turn the carousel. Button 4 chooses. Button 5 takes 10 s off the countdown. A guest or first-time player cannot choose FREEDOM. A first-time player cannot choose DAN COURSE. After the choice the game charges the cost, which is zero on free play, and fixes the number of tracks.

6.5 Online-only screens

Next come screens that need the player's online profile: rankings, circle results, the login bonus, area select, tour members, presents, tickets and a tutorial offer. A guest passes through them almost at once.

6.6 Music select

Music select is where the player picks a song, a difficulty and the options. On the first track it opens on the category cards. On later tracks it opens on the song wheel, at the last song played. It works in six stages, listed in Table 34.

Table 34. The stages of music select and their main controls.

Stage	Main controls
Category	button 4 enters; buttons 3 / 6 change category; buttons 1 / 8 change difficulty; SELECT opens the sort settings
Song	button 4 chooses; button 5 goes back; buttons 3 / 6 scroll; buttons 2 and 7 together switch between the DX and STANDARD chart
Difficulty	button 4 chooses; buttons 3 / 6 pick a harder or easier chart
Play settings	TRACK START, options, headphone volume, in-store matching
Options	buttons 3 / 6 move between items; touch keys change values; button 4 confirms
Sort settings	tab layout, song order and play-status display

The song wheel shows nine cards, with the centre card at full size and its neighbours at 70 %. A short preview of the centred song plays once the wheel stops. Each action briefly locks input, from 50 ms after a difficulty change to 1200 ms after a choice. After TRACK START the game waits 1000 ms, then wipes to track start.

6.7 Track start and play

The track-start screen waits 1.0 s, then plays a 3.33 s animation. A "TRACK n" plate appears over a scrolling town, and the song card pops in. The card shows the player's best achievement, rank and DX score for the chart. Achievement is the score as a percentage, with 101 % as the maximum. The DX score counts judgement quality, at most three points per note.

The play screen starts in three steps:

1. It waits until the music, the background movie and the notes are loaded.
2. It holds 1000 ms, then starts the note clock.
3. 91 ms later it starts the music and movie. The lead makes up for the delay of the sound output.

Play ends when every note has been judged. The screen then holds until the song has ended and 6 s have passed. Two button chords end play early. Holding buttons 2, 3, 6 and 7 for 3.5 s skips the track, if the track-skip option allows it. Holding buttons 3, 4, 5 and 6 for 2.5 s restarts the song, in FREEDOM mode only.

6.8 Result and the end of the credit

The result screen stages the score in about 4 s, with the partner calling the rank at 1.02 s. Button 4 then reads NEXT.

The DX rating is the player's skill rating, rebuilt from their best scores. It is recalculated after every track, and a guest's rating starts from zero. Its plate changes colour at the thresholds in Table 35.

Table 35. DX rating plate colours.

From rating	Plate	From rating	Plate
0	White	12000	Bronze
1000	Blue	13000	Silver
2000	Green	14000	Gold
4000	Yellow	14500	Platinum
7000	Red	15000	Rainbow
10000	Purple		

If tracks remain, a 0.47 s wipe leads back to music select. Otherwise the credit ends:

1. Online players see unlocks, the area result and new items.
2. The save screen uploads the results, taking at least 2.0 s.
3. If the operator allows it, the continue screen offers another credit.
4. The game-over screen plays for 6.7 s while the town sky turns to night.
5. The cabinet returns to the attract loop.

Every screen change uses the same wipe. The panels close in 0.467 s, the next screen loads behind them, and the panels open in 0.417 s.

Figure 11 shows the screens of one guest credit, from the title to play.



Figure 11. Screens of the game during one guest credit: (a) title, (b) entry, (c) attention, (d) mode select, (e) music select, category cards, (f) music select, song wheel, (g) music select, play settings, (h) play. The lower strip of each tile is the upper monitor.

6.9 Monitors, credits and players

The game draws the round main monitor and the upper monitor as one 1080 × 1920 portrait picture. A 1080 × 1080 square feeds the main monitor and a 1080 × 450 strip feeds the upper one. The upper monitor shows the player's name plate, rating and track counter, and the song during play.

The cabinet knows two game prices and a separate FREEDOM price. On free play every payment succeeds at no cost, and any button starts a credit.

Two players can share a cabinet, one per side. Each gets a full copy of every screen. Two-player features, such as the SYNC count of how closely the players match, appear only when both are in.

6.10 Tracks per credit

A track is one song within a credit. The game fixes the number of tracks when the mode is chosen, as Table 36 shows.

Table 36. Tracks per credit.

Situation	Tracks
NORMAL, one player	3
NORMAL, two players, or an extra track bought at mode select	4
A player's very first credit, alone	1
DAN COURSE	4
KALEIDXSCOPE	3 (1 at the final gate)
Event mode, set by the operator	as set

A long song counts as two tracks. The game will not offer one on the last track.

FREEDOM mode replaces the track count with a time budget. With T the budget in seconds:

$$T = 600 + 120 \times \text{pass bonus}$$

The pass bonus is 1 when the player's DX pass carries the extra-time effect, else 0. A pass with the effect therefore gives 720 s, or 12 minutes. The clock pauses during the wipe between tracks. When time runs out, the current song is the last.

6.11 Course, KALEIDXSCOPE and utage

DAN COURSE. The player plays four set songs in a row against a life gauge. Each course sets the starting life, the life regained per song, and the life lost per Great, Good and Miss. The game ships 294 courses in three kinds: dan certification, true dan certification and random dan certification.

KALEIDXSCOPE. This mode is a set of ten gates:

- six coloured doors: blue, white, purple, black, yellow and red;
- a central tower;
- a boss gate;
- a "door of hope";
- a final boss gate.

Each gate needs a key and has a three-song course and a reward song.

Utage. Utage (宴, "party") charts are joke and special charts in their own category. Some fix the options, and some are played by two players together.

6.12 Player options

The option screen has five pages: speed, game, display, design and sound. Three presets (Basic, Advanced and Expert) set the tap and touch speed to 2.0, 3.0 and 4.5. Only the Custom preset uses the player's own speeds. Table 37 lists the options a player is most likely to change.

Table 37. Main player options.

Option	Values	Default
Tap speed, touch speed	1.00 to 10.00 in steps of 0.25, and SONIC	4.50
Slide display timing	−1.0 to +1.0 in steps of 0.1	0.0
Judgement timing A and B	−2.0 to +2.0 in steps of 0.1	0.0
Track skip	Off, hold button, automatic at a target (S, SS, SSS, best, rival), or at a life total	Off
Mirror	Off, left–right, up–down, rotate 180°	Off
Movie brightness	five steps, Dark to Bright	Normal
Centre display	Off, combo, achievement, borders, DX score (11 choices)	Combo
Judgement display	15 layouts	Type 1-A
Tap, hold and slide design	Simple, Classic and others	Simple
Judgement-line design	18, one for each game version	CiRCLE
Answer sound volume	Off, 1 to 10	6
Other sound volumes	Off, 1 to 5	5
Tap, break, slide and EX sound sets	8, 50, 50 and 7 sets	Default
Headphone volume	1 to 20	1

The speed option is stored as a speed value v. With s the speed shown to the player:

$$v = 100\,(s + 1)$$

For example, a speed of 4.50 is stored as $100 \times 5.5 = 550$. Speed 1.00 is 200 and speed 10.00 is 1100. SONIC is a fixed value of 5000, at which notes appear for an instant. A higher value means notes take less time to reach the ring.

6.13 Categories and sorts

The tabs above the song wheel follow one of six layouts, chosen in the sort settings. Table 38 lists them.

Table 38. Tab layouts in music select.

Layout	Tabs
Genre	22: eight genres (POPS & ANIME, niconico & VOCALOID, Touhou Project, GAME & VARIETY, maimai, ONGEKI & CHUNITHM, utage, "strong") and automatic folders such as new songs, favorites and ALL
Song title	16 rows of kana and the alphabet
Level	25, from level 0 to 15+
Version	26, from the first maimai to CiRCLE
Rank	14 clear ranks
All songs	1

Within a tab, songs can be ordered by recommendation, title, level, date added, BPM, rank, full-combo icon, SYNC icon or DX score. The game offers 1,500 songs: 583 with STANDARD charts, 847 with DX charts and 70 utage entries. A song with both kinds of chart appears once.

6.14 Collections

Players collect cosmetic items that decorate their name plate and profile. Table 39 gives the counts the game ships. Each item has an unlock condition and a short text that explains it.

Table 39. Collectible items.

Collection	Items
Titles	3,949
Icons	1,449
Name plates	480
Frames	502
Partners	26
Tour characters	1,274

Tour characters are a separate system. Up to five of them form the player's touring team, which moves the player along areas (see "Areas"). They level up with play and "awaken" at levels 9, 49, 99, 299, 999 and 9999.

Figure 12 groups all these systems and gives their sizes.

Progression and collection systems, with the number of entries shipped in version 1.60

Songs and categories					
Songs	1,500	Genres	22	Versions	26
Levels	25	Title rows	16	Clear ranks	14
Collections					
Titles	3,949	Icons	1,449	Name plates	480
Frames	502	Partners	26	Closeness rewards	144
Collection tabs	224	Photo frames	19		
Tour characters and cards					
Tour characters	1,274	Character tabs	208	Awakening steps	6
Card Maker cards	282	Character cards	310	Card packs	130
Progression					
Areas	202	Area treasures	2,527	Bonus song lists	221
Login bonuses	68	Missions	6	Tickets	27
Shop items	197	Events	1,345	Information pages	11
Modes and competition					
Dan courses	294	KALEIDXSCOPE gates	10	KALEIDXSCOPE song groups	28
PERFECT CHALLENGEs	48	Friend-battle classes	26	Friend-battle seasons	26
Season rewards	219	Festa teams	3	Circle reward months	7

Figure 12. The game's progression and collection systems grouped by purpose, with the number of entries of each that version 1.60 ships.

6.15 Partners and voices

A partner is the character who guides the player. The partner appears on the result screen and speaks throughout the credit. There are 26 partners. Two are owned from the start: Derakkuma and Lime-kuma & Lemon-kuma. A new profile has Derakkuma.

Voices. The game has a system voice set of 150 lines and, per partner, a set with the same 94 line slots. It loads the equipped partner's set on the attention screen, so each partner voices the same moments differently. The partner speaks in music select, calls the rank from A to SSS+, and asks the continue question. Another 475 cues cover sound effects and music.

Closeness. Closeness measures a player's bond with a partner. It rises by one for each present the player gives that partner. With C the closeness and g the presents given:

$$C_{\text{new}} = C + g, \qquad g \le \text{presents held}$$

A player holds at most 9,999 presents. Presents come from the shop, from circle rewards and from area treasures. Each partner has rewards at closeness 10, 50, 200, 300, 500 and 1000. For example, a player at closeness 180 who gives 25 presents reaches 205 and collects the reward at 200.

6.16 Areas

Areas are paths the player walks by playing. The game has 202 areas on seven islands, one island per version. Treasures along each path give tour characters, songs, plates, frames and titles. The game adds the distance once per credit, after the last track.

Distance formula. The distance D added in one credit is:

$$D = \left\lceil \left(\sum_{i=1}^{5} 1000\, m_i + B_{\text{play}} + B_{\text{song}} \right) \times n \times t \right\rceil_{1000}$$

The terms are:

- m_i is the step value of tour member i: 4 for the leader, 2 for a member whose colour matches the area, and 1 for anyone else. On an endless area, a reborn member adds 1 more.
- B_{play} is a bonus for the best result of the credit: 4000 for any FULL SYNC, 3000 for ALL PERFECT, 2000 for FULL COMBO, and 1000 for a two-player play.
- B_{song} is the area's bonus when any song of the credit is on its bonus list. It is 2000 in the first area, and 1000 on an endless area without a list.
- n is the number of tracks, counting a long song twice.
- t is the ticket multiplier (Table 40), or 1 without a ticket.
- $\lceil \cdot \rceil_{1000}$ rounds up to the next multiple of 1000.

Any distance the player had stored is then added to D. Section 9 restates this formula with tickets written as a percentage.

Worked example. A team has a leader (4), two colour-matched members (2 each) and two others (1 each). The character sum is $1000 \times (4 + 2 + 2 + 1 + 1) = 10\,000$. The player gets a FULL COMBO, which adds 2000, and plays a bonus song, which adds 2000. Over three tracks with a two-credit ticket:

$$D = (10\,000 + 2000 + 2000) \times 3 \times 3 = 126\,000$$

6.17 Tickets

A ticket is bought or earned before music select and raises the rewards of one credit. Paid tickets cost extra credits. A paid ticket costing k extra credits multiplies three things by the same factor:

$$t = k + 1$$

The factor applies to area distance, tour-member level gains and festa points. Table 40 lists the ticket kinds.

Table 40. Ticket multipliers.

Ticket	Extra credits	Distance	Tour-member levels	Festa points
Paid	1 to 5	×2 to ×6	×2 to ×6	×2 to ×6
Event or shop (single use)	0	×1.5, ×2, ×3 or ×5	×1	×1
Event or shop (up to 5 uses)	0	×1 to ×5	×1.5 (one kind ×5)	×1
Long-running event (up to 99 uses)	0	×10 or ×20	×1.5	×1

6.18 Login bonus, missions and events

Login bonus. A login bonus is a stamp card that gains one stamp per day played. The game ships 68 cards. Most need 10 stamps, a few need 5 or 3. Each card ends in one reward, most often a partner or a new song.

Missions. Missions are weekly goals of three kinds: logging in, playing songs, and playing with a partner player. Each kind has two levels. They pay 100 to 800 maimai miles, the in-game currency spent in a shop of 197 items.

Events. Almost all content belongs to one of 1,345 numbered events. The online service decides which events are open, so content unlocks on a schedule without an update. Only one event is always open.

6.19 Circles

A circle is a club of players, new in the CiRCLE version. Members earn circle points each month. The game computes the points each credit earns with the formula in Section 9; the online service keeps the monthly total. At the start of a new month, each member collects every reward step that their circle's points for the last month reached. A month has 15 steps, from 500 to 10,000 points. With P the last month's points and p_k the threshold of step k, the rewards are the steps with:

$$p_k \le P$$

For example, 2,300 points reach the steps at 500, 1000, 1500 and 2000. They give 200 miles, a ticket, three presents and a second ticket. The game ships reward tracks for seven months, September 2025 to March 2026.

6.20 Festa

A festa is a season in which players join one of three teams. The shipped season's teams are Raz, Salt and Chiffon. Points are added after each credit. With c the cleared tracks, ℓ the cleared long songs, and t the ticket factor:

$$F = \left(100\,(c + \ell) + R_{\text{rank}} + 200\,[\text{two players}] + 500\,[\text{first credit today}]\right) \times t$$

The terms are:

- A long song counts twice through ℓ.
- R_{rank} is a rally bonus paid during rally periods: 0 for the leading team, 100 for the second and 200 for the third. It helps teams that are behind.
- A bracket is 1 when its condition holds, else 0. The 500-point daily bonus is paid once per day.

Worked example. A player clears one normal song and one long song, so $c = 2$ and $\ell = 1$. Their team is second in a rally. They play alone, it is their first credit of the day, and they use a one-credit ticket ($t = 2$):

$$F = (300 + 100 + 0 + 500) \times 2 = 1800$$

The season reward, a new song, unlocks at 20,000 points.

6.21 Competition

Friend battles rank two-player matches in 26 classes from B5 upward, over 26 seasons. PERFECT CHALLENGE offers 48 songs played with a small life total that grows as days pass.

6.22 Language

The game is a Japanese release and ships no other language. Menus, item texts and voices are Japanese, and tabs and buttons are drawn artwork. The English names in this paper are translations.

7. The cabinet online

In brief. A maimai DX cabinet depends on a small set of SEGA services reached through ALL.Net, SEGA's arcade network. Two facts matter most. First, the game itself talks only to the title server, which stores player profiles; a separate cabinet service handles everything else. Second, every title-server request is a compressed and encrypted JSON record, and a player's whole profile is saved in one request at the end of each credit.

7.1 The ALL.Net server infrastructure

Figure 13 shows the pieces and how they connect. Two programs run on the cabinet PC. The **game program** plays maimai DX. The **cabinet service** is SEGA's common arcade background program. It owns the card reader, the coin and credit counter, the network settings and every connection to SEGA except the title server. The game asks the cabinet service for results and never opens those connections itself.

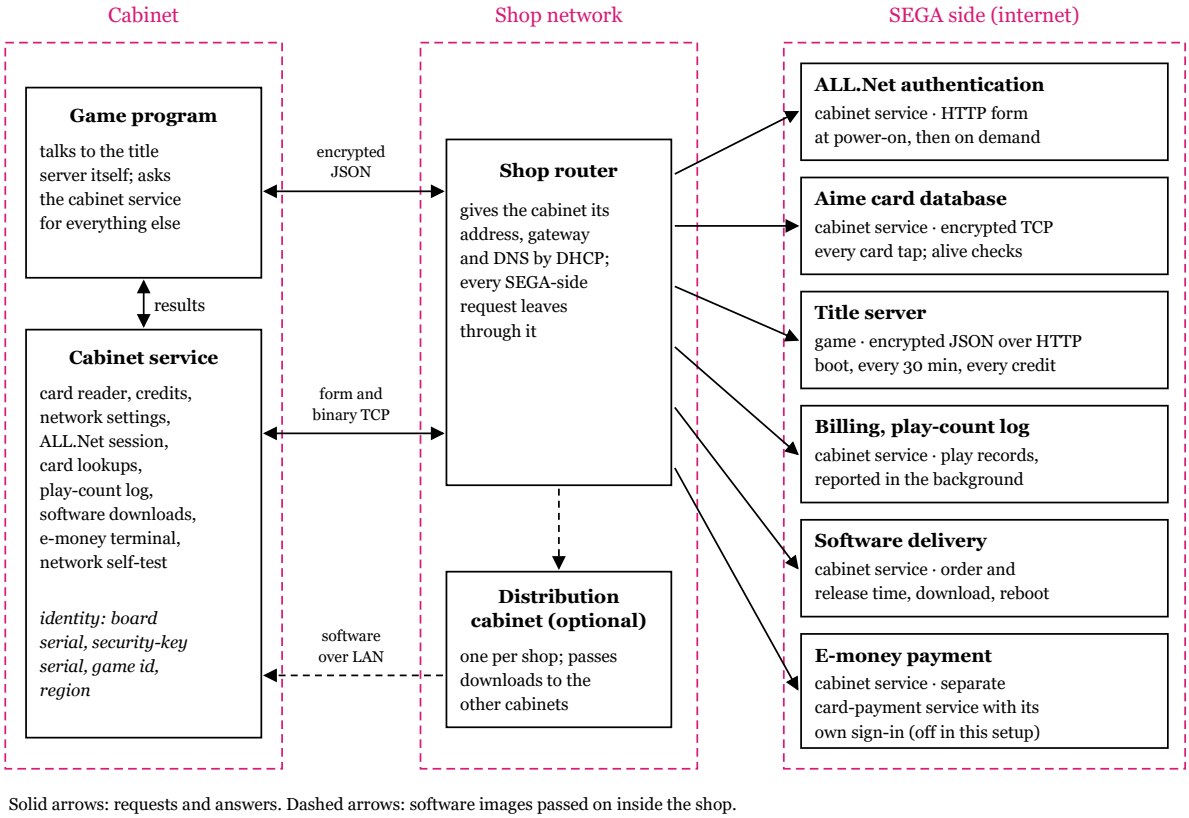


Figure 13. The services a cabinet relies on: the game and the cabinet service inside the cabinet, the shop router and an optional distribution cabinet on the shop network, and six SEGA-side services, each labelled with who contacts it, how and when.

Table 41 summarises each service. The subsections after it explain the ones a reader needs in more detail.

Table 41. The services a cabinet uses, who contacts them, and what happens when they fail.

Service	What it does	Contacted by	Protocol	When	If unreachable
ALL.Net authentication	Confirms the cabinet is licensed and tells it where the title server is	Cabinet service	HTTP form request	At power-on	Game runs offline; guests can play, card players cannot log in
Aime card database	Turns a card's access code into a player number	Cabinet service	Encrypted binary messages over TCP	At every card tap, plus regular alive checks	Card login is disabled; the network icon shows offline
Title server	Stores profiles, scores and cabinet-wide game data	Game	Encrypted JSON over HTTP or HTTPS	At boot, every 30 minutes, during every credit	Card login is disabled; the game retries on a back-off schedule
Billing and play-count log	Records each paid or free play for SEGA's accounts	Cabinet service	Background reports	Each play is logged; reports go out in the background	While a report is being sent, new logins wait; the exact rule for a full log is not known
Software delivery	Downloads new game and option images on a schedule	Cabinet service	Download instructions plus file transfer	Checked after authentication	The cabinet keeps its current version
Shop distribution cabinet	Passes downloaded software to the other cabinets in the shop	Cabinet service	Shop LAN	At start-up and at a set interval	Other cabinets keep their current version
E-money payment	Card payment for credits through a third-party service	Cabinet service	Its own service, signed in separately	When a player pays	Paying by e-money is unavailable; coins still work

7.2 ALL.Net authentication

ALL.Net is the first service the cabinet contacts. The cabinet service sends one form request at power-on. It names the game, its version, the cabinet's serial number, its IP address, its firmware and boot versions, and the number of network hops to the server. The reply confirms the cabinet and carries:

- the title server's base address;
- the shop's id, name and nickname;

- the region code and up to four region names, and the country;
- the current time and the shop's time zone;
- an access string that the game sends with its own cabinet-level requests;
- a list of certificate fingerprints for the title server.

The cabinet accepts broadband and DSL lines. Mobile lines are switched off in this configuration. A separate wireless-router option, called ALL.Net Air, also exists and has its own status checks.

7.3 Software delivery and the distribution cabinet

SEGA delivers updates as two kinds of image: the game program and optional data. Each image has an **order time**, when the download is announced, and a **release time**, when the cabinet may switch to it. The cabinet service downloads in the background. Test mode shows the current version, the download progress and both times.

A shop can mark one cabinet as its **distribution cabinet** with a DIP switch. Only that cabinet downloads from SEGA. The others in the shop fetch the images from it over the shop LAN. At start-up the game checks this link. If it finds two distribution cabinets on one LAN, it reports a duplicate error.

When an update is ready to install, the game stops accepting logins and reboots between credits. It also reboots during a daily maintenance window that the title server sets, and after a reboot interval that the title server can also set.

7.4 The shop network and the network test

The cabinet gets its address, gateway and DNS server from the shop router by DHCP. The configuration requires the cabinet to sit behind that router rather than connect to the internet directly. The **hop count** is the number of routers between the cabinet and ALL.Net. The cabinet reports it at power-on and shows it in the network test.

The game runs the same network test at power-on and in test mode. It checks ten items in a fixed order and marks each one Good, Bad or N/A. The IP address check fails if it takes longer than 15 seconds. After the ten items the game checks the title server.

Table 42. The items of the network test, in order.

Item	What it checks
IP address	The cabinet has an address
Gateway	The shop router answers
LAN DNS	Names can be looked up
Hops	Number of routers to ALL.Net (shown as a number)
Line type	Broadband, DSL or mobile
ALL.Net authentication	The power-on handshake succeeded
Aime	The card reader and the card database work
E-money	The e-money service is available
Aime Pay	The card-payment service is available
ALL.Net Air	The wireless router, if one is fitted
Title server	Shown after the ten items: the game's own server answered

7.5 What the game does when a service is down

The network icon on screen shows **online** only when three things hold. The title server answers, the Aime card database answers, and the cabinet has downloaded the game data at least once since power-on. Otherwise the icon shows offline.

When the title server stops answering, the game retries on a back-off schedule. It retries at once, then every 10 seconds for 7 tries, then every minute up to the 66th try, then every 10 minutes. While the server answers, the game refreshes the cabinet-wide data every 30 minutes.

7.6 The cabinet's identity

A cabinet identifies itself with four values held by the cabinet service:

- the **main-board serial**, fixed to the PC board;
- the **security-key serial**, fixed to a plug-in hardware key that licenses the game;
- the **game id**, a four-character code for maimai DX;
- the **region** the key is licensed for.

Each serial exists in a 16-character standard form and an 11-character short form. The cabinet service sends the security-key serial to ALL.Net. The game sends the short form to the title server as the cabinet's client id.

The shop details flow in one direction. ALL.Net sends the shop id, name, nickname, region and country at power-on. The cabinet service passes them to the game, and the game keeps them as the shop's data. The game then sends the shop id and region with every player login. It also stores the last shop, region and country in the player's profile when it saves.

7.7 Identifying a player: the Aime card

An **Aime card** is SEGA's player card; a phone can stand in for one. It carries a 20-digit **access code**. When a player taps the reader, the cabinet service looks the code up in the Aime card database and returns a 32-bit **player number**. If the database cannot be reached, the reader can still return the card's own FeliCa id for offline use.

The lookup uses a small binary protocol over TCP. Each message starts with a 32-byte header. The header names the command, the game, the shop and the security key. The cabinet service pads each message to a multiple of 16 bytes and encrypts it with AES-128. A short session opens with a greeting, sends one lookup, and closes. The cabinet service also sends small log messages for each credit: entering, continuing and leaving.

A guest has no card, so the game invents a player number from the shop's id. The game still uploads a guest's play under that number.

7.8 Talking to the title server

The game sends every title-server request as one HTTP POST. The body is one JSON record, and the answer is one JSON record. Figure 14 shows how a request travels. The game proceeds in four steps:

1. It writes the request as JSON text.
2. It compresses the text with zlib.
3. It encrypts the result with AES-256-CBC, using a small native library that ships with the game.
4. It posts the result to a hashed address under the title server.

The hashed address hides the request's plain name. It is a 32-character hexadecimal hash of the name combined with a fixed value that changes with each game version. The response returns the same way. The game decrypts it, decompresses it and reads the JSON back.

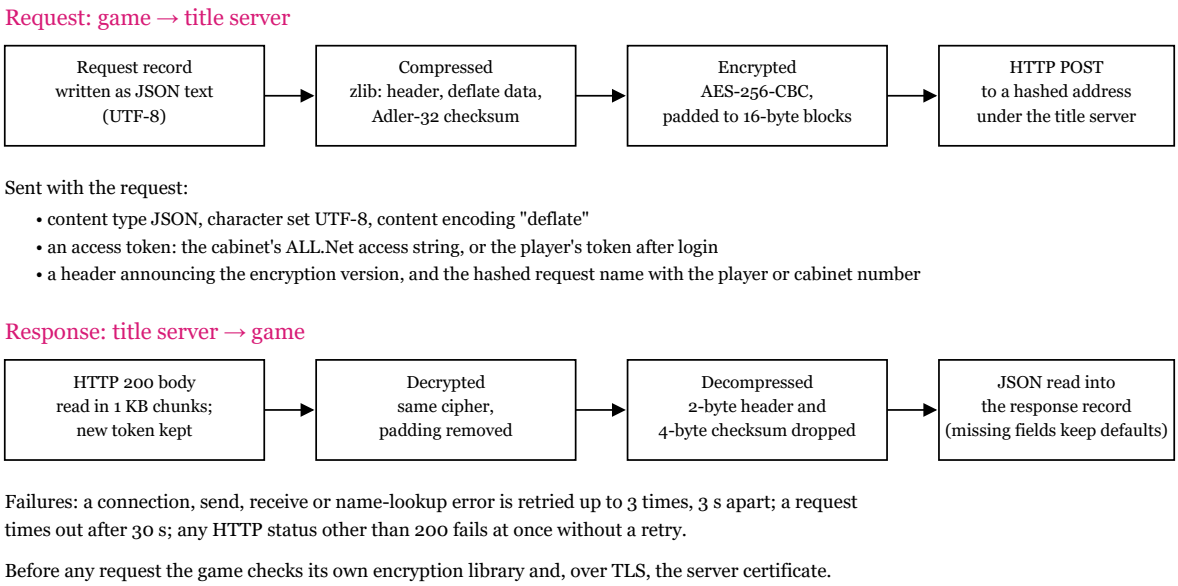


Figure 14. How a request reaches the title server and how the answer comes back: JSON, zlib compression, AES-256-CBC encryption and an HTTP POST, then the same steps in reverse.

There are 66 kinds of request. Figure 15 groups them by what they carry.

The 66 kinds of title-server request, by what they carry

Group	Count	What the requests do
Session	5	log a player in or out, issue or drop a guest token, ping
Cabinet-wide game data	11	settings and maintenance window, events, rankings, tournaments, tickets, banned songs, weekly missions, KALEIDXSCOPE, festivals, shop circle
Cabinet reports	5	shop settings, bookkeeping, play time, error logs, test-mode settings
Player profile	9	preview, main profile, extended profile, options, rating, activity, regions played in, cards, portrait
Collection	11	items by kind, characters, favourites, new items, shop stock, tickets, partner closeness, login bonuses, maps
Scores and progress	8	song scores, courses, ghosts, score rankings, recommendations, missions, gates
Friends, rivals, circles	12	friend check and registration, rival scores, seasons, circles, festivals
End-of-credit saves	5	play record per track, portrait, photo, ticket purchase log, save-everything

Figure 15. The 66 kinds of title-server request grouped by what they carry, with the number in each group.

Each request also carries an **access token**. Before login it is the cabinet's ALL.Net access string. After login it is a personal token that the title server issues to the player. Table 43 lists the timing rules.

Table 43. Title-server timing rules.

Rule	Value
Request timeout	30 seconds
Retries after a connection error	Up to 3
Wait between retries	3 seconds
HTTP status other than 200	Fails at once, no retry
Refresh of cabinet-wide data while online	Every 30 minutes

7.9 Integrity and certificate checks

The game checks itself before it sends anything. It computes a SHA-256 digest of its own encryption library. It compares the digest with a value built into the program. If they differ, the game blanks every request address, so no request can reach the server.

Over HTTPS the game also checks the server's certificate. It accepts only a certificate from SEGA's title-server domain whose fingerprint matches one sent by ALL.Net at power-on. ALL.Net thus vouches for the title server, not the usual certificate authorities.

7.10 The course of a credit

Figure 16 shows one credit from power-on to logout.

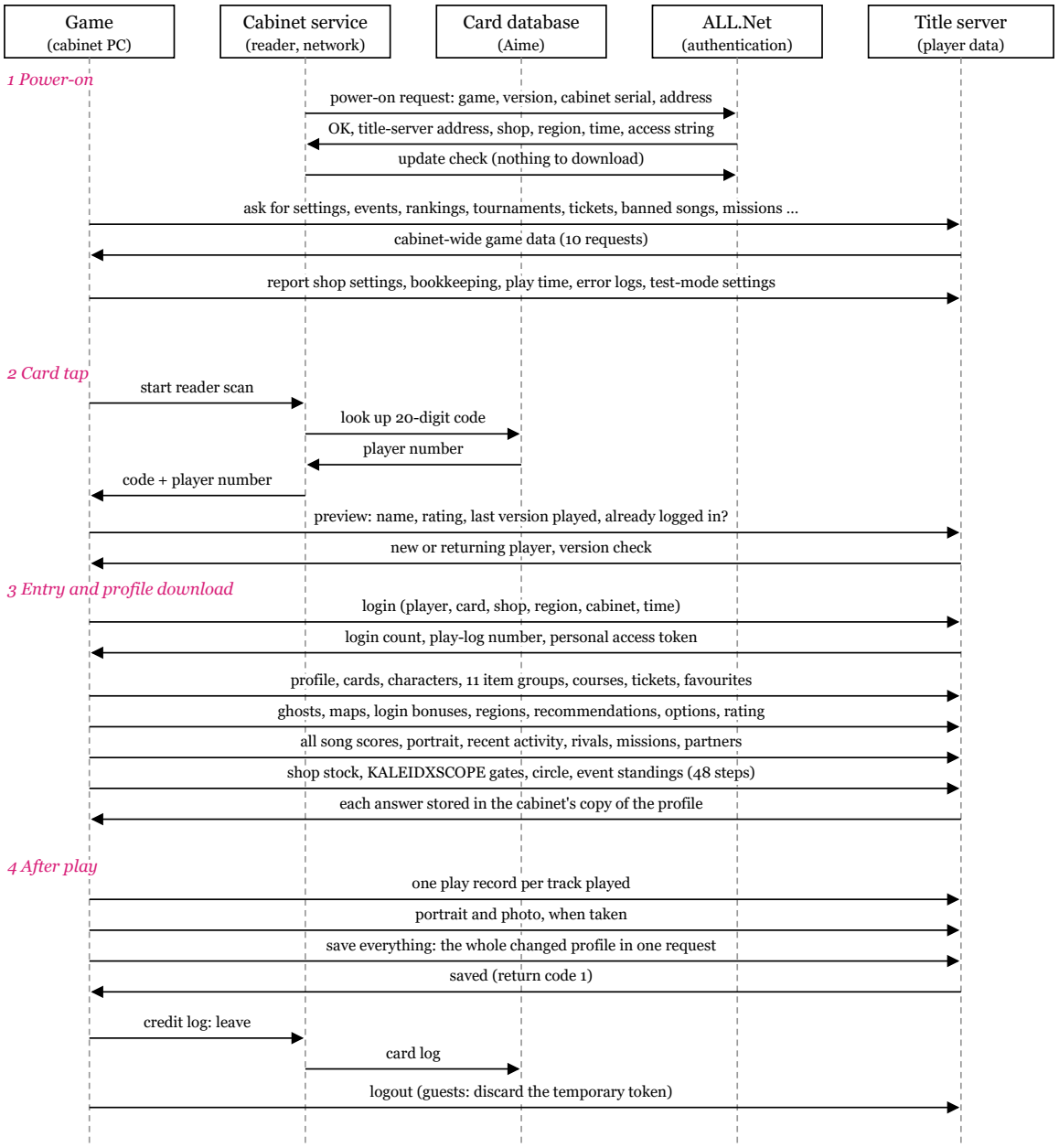


Figure 16. One credit on the network: power-on authentication and the cabinet-wide downloads, the card lookup, the preview, login and 48-step profile download, and the uploads that end with saving the whole profile and logging out.

1. **At boot.** The game downloads the cabinet-wide data in ten requests. These cover the game settings, rankings, events, tournaments, ticket offers, withdrawn songs, weekly data, the KALEIDXSCOPE schedule, festivals and the shop's circle. The settings include a maintenance flag and the daily reboot window. The game then reports the shop's settings, bookkeeping, play time, error logs and test-mode settings.
2. **Preview.** After a card tap the game asks for a preview of the player. The preview gives the name, rating, chosen icon and name plate, and the last version played. It also says whether the

player is already logged in elsewhere. A player with no name or no last-play date is new. A player already logged in elsewhere is refused.

3. **Login.** The game logs the player in with the player number, access code, shop, region and cabinet. The reply gives the login count, a play-log number for this credit and the player's access token.
4. **Profile download.** The game fetches the profile in 48 fixed steps, one request each. It loads the main profile first, then the collection, options, rating, all song scores, social data and event standings. Item lists arrive one kind at a time and in pages.
5. **Uploads.** After play the game sends one play record per track and the portrait and photo, if any.
6. **Save.** The game sends the whole changed profile in one request. The server answers with a single return code. The game treats anything but success as a failed save.
7. **Logout.** The game logs the player out. For a guest it discards the temporary token instead. If a credit is abandoned, the game logs out both sides at once.

7.11 What a saved profile contains

The final save carries the profile in 29 parts. Three parts are single records: the main profile, the extended profile and the options. Most other parts are lists. For a list, the game sends only the entries that changed during the credit and marks each as new or updated. Table 44 summarises the parts.

Table 44. The main parts of a saved profile.

Part	What it holds
Main profile	Name, icon, name plate, title, partner and frame; ratings, class and course ranks; play and win counts; totals of DX score, sync and achievement per difficulty; first and last versions, shops, regions and dates
Extended profile	Last song and difficulty chosen, music-select position and sorting, result-screen settings, photo consent
Options	48 settings: note, slide and touch speeds, note designs and sizes, display and judgement options, every sound effect and volume
Rating	The rating and the songs that count toward it, from older and from the newest versions
Song scores	Per song and difficulty: play count, achievement, combo and sync flags, best DX score, rank
Collection	Characters with level and awakening; items by kind with stock; favourites; ticket stock; partner closeness
Progress	Maps, login bonuses, courses, KALEIDXSCOPE gates, missions, recent activity, ghost
Play records	One record per credit with mode, credits, tracks and flags
Social	Friend-season ranking, circle membership and points, festival results

Two of the main-profile values are 64-bit unlock flags. A server must store them as exact integers, because they exceed what a double-precision number holds.

8. Conclusion

In brief. maimai DX runs on a few exact rules, applied the same way every time. The cabinet around them is two programs working together, linked to a small set of SEGA servers.

8.1 The rules

Time is a grid of 384 steps per bar. Every note is judged against a fixed window table, measured in sixtieths of a second. Each judgement is worth a fixed number of points.

Achievement is those points as a share of the maximum. Break notes fill a separate 1% pool, which is why a perfect play reads 101%. Rank and rating are both lookups on achievement. A chart's rating multiplies its internal difficulty constant by the achievement and by a step coefficient. A player's rating adds up two lists of best charts: one for the newest songs and one for everything older.

8.2 The cabinet

The game draws the screens and applies the rules. A separate cabinet service owns the hardware, the card reader, the credits and the network. The touch panel, the buttons and the lights are simple serial devices that the game reads every frame.

8.3 The network

Behind the cabinet stand a few SEGA services. ALL.Net signs the cabinet in when it powers on, delivers updates and keeps the play log. The Aime database turns a card into a player number. The title server stores every player's profile. The game talks to the title server with compressed, encrypted JSON requests. It loads a profile in groups when a credit starts, and writes the whole profile back in one request when the credit ends.

8.4 What is left

The less exact parts are the ones the player sees least: what the hardware does inside its own boards, and how the servers treat what the cabinet sends. Everything between a touch on the glass and the number on the result screen follows the rules in Sections 4 and 5, and the formulas in Section 9.

9. Formula reference

In brief. This appendix collects every formula the game uses to turn play into a number the player sees. Each entry gives the formula, what its symbols mean and one worked example with real values. The scoring formulas repeat those of the scoring section; the others cover timing, note speed, the map, tour members, partners, circles, festa and login bonuses.

9.1 Chart timing

Chart position to milliseconds. A chart places notes on a grid of ticks, 384 per bar of four beats. The game converts a tick position to a time by walking the tempo changes before it:

$$t = t_{\text{adj}} + \sum_{\text{segments}} \frac{\Delta n}{384} \times 4 \times \frac{60\,000}{\text{BPM}}$$

where t is the note time in ms, Δn is the number of ticks spent in a segment of constant tempo, BPM is that segment's tempo, and t_{adj} is the note-timing adjustment below.

Example. A note at bar 2, tick 192 of a 150 BPM song lies 2.5 bars in. That is $2.5 \times 4 \times 400 = 4000$ ms, plus 60 ms of default adjustment: 4060 ms.

Note-timing adjustment. The player's timing-adjust option A (0 to 40, default 20) moves the whole chart:

$$t_{\text{adj}} = \frac{A - 20 + 36}{10} \times 16.666666$$

Example. The default $A = 20$ gives $3.6 \times 16.666666 = 60$ ms. The extremes give 26.7 ms and 93.3 ms.

Scroll-speed changes. Some charts change the scroll speed for a stretch of time. A note's drawn time is pushed by each change it passes:

$$\tau = t + \sum_{\text{changes}} \text{overlap} \times (s - 1)$$

where τ is the time used to draw the note, t its real time, overlap is the part of the change that lies between now and the note in ms, and s is the change's speed factor. Judgement always uses t.

Example. A 500 ms stretch at speed 2 between now and the note pushes it $500 \times (2 - 1) = 500$ ms further out, so it appears farther from the ring.

9.2 Note speed

Tap and slide approach. The tap speed option v (1.00 to 10.00 in steps of 0.25, plus SONIC) maps to a beat length:

$$b = \frac{60\,000}{100\,v + 100}, \qquad t_{\text{appear}} = 8b, \qquad t_{\text{travel}} = 4b$$

where b is the internal beat length in ms, t_{appear} is how long before its time a tap's guide appears, and t_{travel} is how long the tap moves from the centre to the ring. The note first grows in place for $4b$, then travels for $4b$. SONIC uses 5000 in place of $100v + 100$.

Example. The default speed 4.50 gives $b = 60\,000/550 = 109.1$ ms. The tap appears 872.7 ms early and travels for 436.4 ms.

Touch approach. The touch speed option u uses its own scale:

$$b_{\text{touch}} = \frac{60\,000}{50\,u + 125}, \qquad t_{\text{appear}} = 5\,b_{\text{touch}}, \qquad t_{\text{close}} = 4\,b_{\text{touch}}$$

Example. The default 4.50 gives $b_{\text{touch}} = 60\,000/350 = 171.4$ ms, so a touch note appears 857.1 ms early.

9.3 Judgement

Window edge. Each window edge is stored in frames and shifted 50 ms late:

$$t_{\text{edge}} = 16.666666 \times (f + o) + 50$$

where f is the edge in frames and o the judge-timing option in frames (−2.0 to +2.0, default 0).

Example. A tap's late Good edge, 9 frames, lies at 200 ms after the chart time, +150 ms from the centre.

Hold release share.

$$p = \min\left(100,\ \left\lceil 100 \times \frac{r}{L - h - 200} \right\rceil\right)$$

where r is the release time, L the hold length and h 100 ms (250 ms for a touch-hold), all in ms. The share selects a row of the hold table.

Example. A 2000 ms hold released for 300 ms gives 18 %, which turns a Critical head into a late Perfect.

Slide Critical allowance.

$$w = T \times \frac{d_{\text{last}}}{d_{\text{total}}}, \qquad -183.3 - \frac{w}{4} \le \Delta \le 283.3 + \frac{w}{4}$$

where T is the star's travel time, $d_{\text{last}}/d_{\text{total}}$ the share of the path in the last hit area, and Δ the player's arrival minus the star's entry into that area.

Example. $T = 1000$ ms and a 20 % last area give $w = 200$ ms and a Critical band from −233.3 ms to +333.3 ms.

9.4 Score

Achievement.

$$A = 100 \times \frac{B}{B_{\max}} + \frac{K}{K_{\max}}$$

where B and $B_{\max}$ are the earned and maximum base points, and K and $K_{\max}$ the earned and maximum break bonus (100 per break).

Example. 400 taps, 100 holds, 50 slides and 20 breaks give $B_{\max} = 425\,000$. One tap Great with everything else Critical scores 100.9765 %.

Stored achievement.

$$a = \lfloor 10\,000 \times A \rfloor$$

Example. 100.9765 % is stored as 1 009 765.

DX score and stars.

$$D = 3n_C + 2n_P + n_G, \qquad q = \left\lfloor 100 \times \frac{D}{3N} \right\rfloor$$

where n_C, n_P, n_G are the Critical, Perfect and Great counts, N the note count and q the percentage that sets the stars (85, 90, 93, 95, 97 % for one to five).

Example. 540 Critical, 25 Perfect and 5 Great on a 570-note chart give $D = 1620 + 50 + 5 = 1675$ and $q = 97$: five stars.

Clear rank.

$$\text{rank}(A) = \max\{\, r : A \geq \theta_r \,\}$$

with thresholds 0, 50, 60, 70, 75, 80, 90, 94, 97, 98, 99, 99.5, 100 and 100.5 % for D up to SSS+.

Example. 98.5 % gives S+.

Combo and chain. The combo C rises by one on any non-Miss judgement and resets on a Miss. In multiplayer the chain sums every player's combo gain each frame:

$$C'_{\text{chain}} = \begin{cases} 0 & \text{if any player missed this frame} \\ C_{\text{chain}} + \sum_i \Delta C_i & \text{otherwise} \end{cases}$$

Example. Two players each hit three notes in a frame: the chain rises by 6.

9.5 Rating

One chart.

$$R_{\text{chart}} = \left\lfloor \frac{c \times \min(a,\ 1\,005\,000) \times k(a)}{10^8} \right\rfloor + b$$

where c is ten times the chart constant, a the stored achievement, $k(a)$ the coefficient (224 at SSS+, 216 at SSS, 208 at SS, 200 at S) and b the ALL PERFECT bonus of 1.

Example. 14.3 at 100.5000 % gives 321.

Total.

$$R_{\text{total}} = \sum_{i=1}^{35} R_i^{\text{old}} + \sum_{j=1}^{15} R_j^{\text{new}}$$

where the old list holds the 35 best charts of earlier versions and the new list the 15 best of the previous and current version.

Example. Averages of 280 and 300 give 14 300, a Gold plate (Gold from 14 000, Platinum from 14 500, Rainbow from 15 000).

9.6 Life

Damage per frame.

$$L' = \max\left(0,\; L - (n_P\, d_P + n_G\, d_G + n_g\, d_g + n_M\, d_M)\right)$$

where n are the new judgements of each kind and d their damage. Perfect Challenge and normal KALEIDXSCOPE gates use $d = 0, 1, 1, 1$. KALEIDXSCOPE EX life uses $1, 10, 20, 50$ and keeps life between 0 and 999. Courses set their own values.

Example. An EX-life phase with 200 life takes 30 Perfects, 4 Greats and 1 Miss: $200 - (30 + 40 + 50) = 80$.

Course recovery.

$$L_{\text{next}} = \min(L + R,\; L_0)$$

Example. A 初段 course ($L_0 = 350$, $R = 20$) at 300 life after a track continues at 320.

9.7 Class ladder

Class value.

$$V = 10\,000 \times i + P, \qquad P' = P + \delta$$

where i is the class index (0 for B5 up to 25 for LEGEND), P the class points and δ the win or loss value. The player moves up after reaching the border and beating the class boss, and moves down when P' falls below 0.

Example. An SS4 player (index 16) with 20 points stores $V = 160\,020$. A loss to a lower class costs 3 points.

9.8 Credits and tracks

Tracks per credit. The game sets the number of tracks per credit by mode. A long song uses two tracks:

$$n_{\text{tracks}} = n_{\text{mode}} - n_{\text{long}}$$

where n_{mode} is 3 for a standard credit, 1 for a single new player, 4 with the extra-track option or in a course, 3 in KALEIDXSCOPE (1 at its last gate), or the operator's event setting; n_{long} is the number of long songs already played.

Example. A standard credit with one long song first plays that song plus one more: $3 - 1 = 2$ songs.

9.9 Map

Distance per credit. At the end of the credit the player's area moves forward:

$$D = \text{roundup}_{1000}\Big((M + G + S) \times n \times \frac{p}{100}\Big) + D_{\text{stock}}$$

where:

- $M = \sum 1000\, m_k$ over the tour members, with $m_k = 4$ for the leader, 2 for a member whose colour matches the area, 1 otherwise, and one more on an endless area for a reborn member;
- G is the medal bonus: 4000 for any FULL SYNC, else 3000 for ALL PERFECT or better, else 2000 for FULL COMBO or better, else 1000 for SYNC PLAY or a trial play, else 0;
- S is 2000 if the area's bonus song was played (1000 on an endless area without one), else 0;
- n is the number of track slots used, long songs counting twice;
- p is the ticket's area percent (100 without a ticket; 150 to 2000 with one);
- roundup_{1000} drops the last digit and rounds up to the next 1000;
- D_{stock} is stored distance carried over from earlier credits (its exact source is not known).

Example. A leader, two matching members and two others give $M = 4000 + 4000 + 2000 = 10\,000$. A FULL COMBO adds 2000 and the bonus song 2000. Over 3 tracks with a 2× ticket: $14\,000 \times 3 \times 2 = 84\,000$.

9.10 Tour members and partners

Tour-member level gain.

$$\Delta\ell = \text{round}_{0.1}(n \times g \times c)$$

where n is the tracks of the credit (in freedom mode the tracks played, at most 6), g is 2 with a Gold Pass and 1 otherwise, c is the ticket's member factor (1 without a ticket, 1.5 for the free ticket, 2 to 6 for paid tickets, 5 for the member level-up ticket), and $\text{round}_{0.1}$ rounds down unless the fraction is 0.1 or more, in which case it rounds up. Awakening stars come at levels 9, 49, 99, 299, 999 and 9999.

Example. 3 tracks with a 1.5× free ticket give 4.5, rounded to 5 levels.

Partner closeness.

$$c' = \min(c + n_{\text{gift}},\ 99\,999)$$

where c is the closeness and n_{gift} the number of presents given. Each present treasure on the map gives one present, and the player can hold at most 9999. Rewards come at closeness 10, 50, 200, 300, 500 and 1000 for each of 24 partners.

Example. Giving 12 presents to a partner at 45 closeness reaches 57 and unlocks the 50 reward.

9.11 Circles, festa and login bonus

Circle points per credit.

$$P_{\text{circle}} = 40\,f + \left\lfloor \frac{3\,s\,\sum_{\text{tracks}} A'}{100} \right\rfloor$$

where f is 1 on the player's first play of the day, s is 2 when both players of a two-player credit belong to the same circle (else 1), and A' is each track's achievement, doubled for a long song and halved for a co-op utage chart. A circle registered to a shop instead gains 1 point per credit.

Example. A first credit of the day with three tracks at 100.5, 99.8 and 98.0 %: $40 + \lfloor 3 \times 298.3/100 \rfloor = 40 + 8 = 48$.

Festa points per credit.

$$P_{\text{festa}} = \Big(\sum_{\text{cleared}} c_{\text{clear}}\,(1 + \ell) + r_{\text{rank}} + b_{2P} + b_{\text{daily}} \Big) \times \max(1,\ m)$$

where c_{clear} is the points per cleared track (100), ℓ is 1 for a long song, r_{rank} is the rally bonus by the team's rank in the shop (0, 100, 200 for 1st to 3rd, during rally periods only), b_{2P} is 200 in a two-player credit, b_{daily} is 500 on the first festa play of the day, and m is the ticket's festa factor (2 to 6 for paid tickets).

Example. Three cleared tracks, the daily bonus and a 2× ticket: $(300 + 500) \times 2 = 1600$.

Login bonus stamps.

$$s' = \min(s + 1,\ s_{\max})$$

where s is the stamp count of the selected bonus card and $s_{\max}$ its size. Of 68 cards, 59 hold 10 stamps, 8 hold 5 and 1 holds 3. The reward arrives when $s' = s_{\max}$.

Example. A card at 9 of 10 completes on the next day's login.